BEESDR — User Manual

A panadapter, DSP receiver and control head for small QRP transceivers that provide I/Q and CAT.

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Author: Gabi Mihaila, YO4WM

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1. What BEESDR is



Figure 1 — BEESDR front panel: main screen on 20 m with the RSS news ticker, the eight-knob encoder unit below, the ATU display (separate unit) on the right.

BEESDR is a self-contained DSP and display unit. It never generates RF itself: on transmit it keys the rig and feeds it processed voice audio, or keys it in CW (§8.11). It takes the **baseband I/Q pair** from a quadrature-sampling detector

(a Tayloe mixer, in practice) and a **CAT serial link** from the same radio, and turns those two into everything a small QRP rig usually cannot afford to have:

- a real **panadapter and waterfall**, 8 / 32 / 48 / 96 kHz of visible spectrum, with a zoom-FFT down to a narrow slice;
- **demodulation in software** — LSB, USB, CW, AM, FM, and the digital-mode passbands — independent of what the rig's own audio chain does;
- **noise tools** that a QRP rig has no room for: an impulse noise blanker, an LMS adaptive noise reducer, an automatic notch, and a frequency-domain (spectral) noise reducer, all switchable individually so they can be compared on the same signal;
- a **calibrated S-meter**, per band;
- a **CW decoder** with an ML classifier;
- **tap-to-tune**: touch a signal on the spectrum or the waterfall and the rig is retuned to it over CAT.

The design intent is explicit: **enhance a QRP transceiver you already own**, rather than replace it. The rig keeps the RF, the filtering, the PA and the transmit chain. BEESDR adds the eyes and the ears.

The reference rig for development is the **uSDX triband** by Barb, WB2CBA (https://antrak.org.tr/blog/usdx-triband-sdr-all-mode-qrp-transceiver/), which brings out both an I/Q pair and a TS-480-compatible CAT port. Anything that does the same will work.

> **Requirements on the host rig**
>
> 1. A baseband **I/Q output** — two audio-level channels, in quadrature, at line level.
> 2. A **CAT serial port** speaking the Kenwood TS-480 dialect (at minimum `FA`, `MD`).
> 3. A common ground and, ideally, a supply rail BEESDR can share.

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2. Hardware



Figure 2 — The enclosure front: 800×480 touch panel, eight-encoder unit, tuning knob, ATU display (separate unit).

Item	Detail
Processor	Espressif **ESP32-S3**, dual core at 240 MHz, with 8 MB PSRAM
Codec	**Wolfson WM8731** stereo audio codec — ADC for I/Q in, DAC for audio out
Codec clock	12.288 MHz crystal (this is what fixes the available sample rates)
Sample rates	**8 / 32 / 48 / 96 kHz**, switched on the fly. This is also the visible span.
Display	**800 × 480** RGB parallel panel, 16-bit colour, backlight on GPIO 45
Touch	**GT911** capacitive controller, I²C on GPIO 48 / 47
CAT	UART1, TX on **GPIO 42**, RX on **GPIO 1**, 9600 / 38400 / 115200 baud
Storage	ESP32 NVS (flash) for all settings, calibration, and colours
Wireless	Wi-Fi for SNTP time; **ESP-NOW** link to a companion XIAO remote
Control knobs (optional)	**M5Stack Unit 8Encoder** — eight push-knobs, a toggle switch and nine RGB LEDs, on the codec's I²C bus (address 0x41), powered from **5 V**. See §8.10.
Toolchain	ESP-IDF 5.5.3, Xtensa GCC

The I/Q pair enters the WM8731 line inputs as a stereo signal — I on one channel, Q on the other. Sideband selection therefore depends on the **sense** of that pair, which BEESDR detects automatically (see the STATUS tab, §8.13) rather than requiring you to get the wiring right first time.

Demodulated audio leaves through the WM8731 DAC.

> The schematic for the interface board — I/Q input conditioning, CAT level translation, supply — is in Annex A.

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3. Software architecture

Everything runs bare-metal on FreeRTOS, with a strict division between the two cores.

Core 0 — the real-time audio pipeline

1. `i2s_aquire_task` reads 1024 stereo I/Q samples from the WM8731.
2. `processing_task` does the work: DC removal → noise blanker → adaptive NR → 1024-point FFT for the display → demodulation (mode dependent) → decimation to the audio rate → filtering and EQ → AGC → back to int16.
3. CAT receive and parsing.

Core 1 — everything the operator sees

`out_to_dac_task` feeds the DAC, and fourteen independent sprite tasks each own one rectangle of the screen: spectrum, waterfall, S-meter, VFOs, CW text, status icons, and so on. Each repaints only when its own data changes, which is what keeps a 800×480 panel looking live on a microcontroller.

Because the two cores are separated this way, a heavy display setting cannot stall the audio, and a heavy DSP setting cannot make the screen tear.

Demodulation: SSB uses Hilbert-transform phasing; CW uses a narrow bandpass centred on the sidetone pitch feeding a Goertzel detector and an ML classifier; AM uses envelope detection; FM uses a discriminator with a noise-derived squelch.

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4. Connecting a transceiver — the uSDX triband example

The uSDX triband (WB2CBA) is the rig BEESDR has been developed and measured against.

Wiring

uSDX	BEESDR	Note
I output	WM8731 LINE IN, left	via the interface board
Q output	WM8731 LINE IN, right	via the interface board
CAT TX	GPIO 1 (RX)	level-shifted on the interface board
CAT RX	GPIO 42 (TX)	idem
GND	GND	one common ground, star point at the rig

First power-up

1. Set the uSDX CAT baud rate and BEESDR's `CAT BAUD` (DSP tab) to the same value. 115200 is the default here.
2. Open the menu → **STATUS**. `MODE / BAND` and `FREQUENCY` should be following the rig within a fraction of a second. If they do not move, the CAT link is not up.
3. On the same tab, `I/Q SENSE` will read **STRAIGHT (I,Q)** or **SWAPPED (Q,I)**. Either is fine — BEESDR compensates. What matters is that it is not stuck on `checking...`.
4. Verify the orientation on the air: tune the rig **down** in frequency and watch a carrier on the spectrum. It should move **right**. If it moves left, press **RE-CHECK I/Q** on the STATUS tab.
5. Calibrate the S-meter for the band you are on — see the CALIB tab.

A known uSDX behaviour worth stating up front

While its own encoder is turning, the uSDX largely stops answering CAT — measured at around 20 replies per second when idle, dropping to 1–2 per second while tuning. The frequency readout therefore lags the knob on the rig, and then catches up when you stop. This is the rig, not the display path. BEESDR deliberately does **not** poll harder to compensate, because asking more often makes it worse: the uSDX parses CAT in the same loop that reads its encoder.

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5. CAT interface

BEESDR speaks the **Kenwood TS-480** CAT dialect over UART1, 8-N-1, at 9600, 38400 or 115200 baud (selected on the DSP tab, applied live).

It is a **client**: it polls the rig and follows it. It sends only one command that changes the rig's state — the frequency set behind tap-to-tune.

Commands BEESDR sends

Command	Rate	Purpose
`FA;`	10 Hz	Get VFO A frequency — the one the operator watches
`FB;`	~0.5 Hz	Get VFO B frequency
`FR;`	~0.5 Hz	Get the receive VFO (0 = A, 1 = B, 2 = memory)
`MD;`	1 Hz	Get the operating mode
`FA` + 11 digits + `;`	on demand	**Set** frequency — tap-to-tune only

Replies BEESDR understands

Reply	Format	Effect
`FA`	`FA` + 11 digits + `;` (14 chars)	VFO A frequency. Becomes the panadapter LO when receiving on A.
`FB`	`FB` + 11 digits + `;` (14 chars)	VFO B frequency. Becomes the LO when receiving on B.
`FR`	`FR` + 1 digit + `;` (4 chars)	Receive VFO. Drives the VFO-A / VFO-B icons and the tap-to-tune target.
`MD`	`MD` + 1 digit + `;` (4 chars)	Operating mode — see the table below.

Mode codes (`MD`)

Code	Mode	BEESDR label
0	—	NONE
1	LSB	LSB
2	USB	USB
3	CW	CW-U
4	FM	FM
5	AM	AM
6	FSK	DIGI-U
7	CW reverse	CW-R
8	TUNE	NONE
9	FSK reverse	DIGI-L

Implementation notes worth knowing

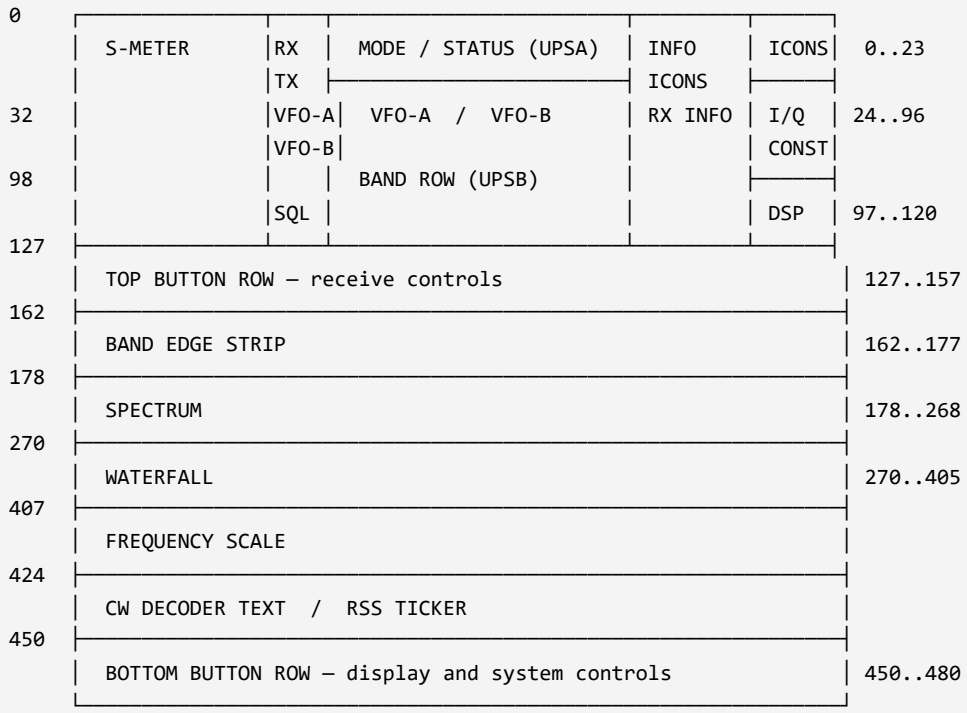
- **Frames are reassembled.** A reply routinely arrives split across two UART reads, and several replies routinely arrive in one. BEESDR accumulates bytes and emits every complete `;`-terminated frame, keeping any partial tail for the next read. Nothing is flushed on a split.
- **Digit fields are validated.** A frame with the right length and prefix but junk in the digit field is rejected rather than painted. The rig is polled continuously, so the next good reply arrives within milliseconds and the display simply holds.
- **Tap-to-tune always addresses** `FA`, even when the rig reports it is receiving on VFO B. On the uSDX, `FA` addresses the *active* VFO and an `FB` set is silently ignored. A build flag (`CAT_SET_FREQ_PER_VFO`) exists for a genuine TS-480, which does implement `FB` as a separate VFO.

- **Poll backoff.** After 5 seconds of complete silence, BEESDR drops to one probe per second. It deliberately does *not* use the 1-second icon timeout for this, because a connected rig goes quiet that long by itself while being tuned.
- **What BEESDR does not do:** it never keys the rig, never sets mode, never sets VFO, and never writes memories.

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6. The screen, element by element

The panel is 800 × 480. From top to bottom:



6.1 S-meter (top left)

An analogue-style needle with a spring-and-damping model, repainted at 15 Hz. It reads S1–S9 and S9+dB, and it is **calibrated per band** — the mapping from raw level to S-number comes from the two capture points you take on the CALIB tab. Uncalibrated, it is indicative only. The foot line carries the CPU die temperature.

Levers: CALIB tab — NOISE REF, SIGNAL REF, CAPTURE NOISE, CAPTURE SIGNAL.

6.2 RX / TX / VFO / SQL column

Five stacked lamps between the S-meter and the mode row.

- **RX** / **TX** — receive and transmit state. Green and red respectively; these two colours are deliberately *not* themeable, because they are a safety convention. While **TX ARM** is on (TX tab, §8.11) the TX lamp has an **orange outline**: the radio may transmit. It turns red while PTT is keyed.
- **VFO-A** / **VFO-B** — which VFO the rig says it is receiving on, from the `FR;` reply.
- **SQL** — the FM squelch. Lit means the gate is **open**, i.e. a signal is holding the channel: it is a *busy* lamp, not a *muted* lamp. It only claims to mean anything in FM with a non-zero level.

Levers: **tap the SQL lamp** to engage/disengage the squelch (FM only — the tap is ignored in every other mode). The level itself is `FM SQUELCH` on the AUDIO tab.

6.3 Mode row (UPSA) and band row (UPSB)

The mode label as reported by CAT (LSB / USB / CW-U / CW-R / AM / FM / DIGI-U / DIGI-L), and the amateur band the current frequency falls in — 160 m through 10 m, IARU Region 1 edges.

6.4 VFO display

VFO-A and VFO-B in large digits, straight from the CAT digit fields. The receive VFO is highlighted.

6.5 Info icons (top right, 615–799 × 0–23)

- **U** — the CAT/UART link. Lit while frames are arriving; goes dark after 1 second of silence.
- **E** — the ESP-NOW link to the companion XIAO remote. The companion also brings the time FT8 needs, and serves decoded charts and FT8 decodes to a browser (§7.4).
- **UTC** — live time from SNTP over Wi-Fi.

6.6 RX info panel (615–717 × 24–123)

Three readouts in one column:

- the **receive passband**, drawn from the live DSP filter edges — not from a table, so it always shows what the audio chain is really doing;
- the **CW tuning aid**, which shows how far the received tone sits from your chosen CW pitch;
- the **volume** bargraph (VOL, 0.05 … 1.00).

The panel keeps updating while the menu or the decoder window is open — it sits above the overlay area, so you can watch the audio react while you change a setting.

6.7 I/Q constellation (726–798 × 24–96)

A 73 × 73 scatter plot of the incoming I/Q pair with a centre crosshair. A clean circle means balanced I and Q; an ellipse means gain imbalance; a tilted ellipse means phase error. It is the fastest way to see that the front end is healthy.

6.8 DSP info (726–798 × 97–120)

A compact `SQL` / `AF` readout — the squelch state and the audio output level. `AF` is in dB, where **0 dB is the point the output clips**. On a clean voice signal the peaks sit around **−6 dB**; if `AF` climbs to −2 … 0 dB you will hear distortion that sounds like reverb or echo — lower `AGC TARGET` or `VOLUME` (see §8.5). Like the RX info panel, it stays live with the menu open.

6.9 Spectrum (178–268)

The panadapter. Full width, 800 px, mapped across the current span (8 / 32 / 48 / 96 kHz divided by the zoom factor).

What can be drawn on it, all independently switchable:

Layer	What it is
Live trace	The current FFT frame, smoothed
Noise floor	A slow per-bin average — the band's own floor
Trail	A decaying trace, so a transient leaves a visible tail
Peak hold	The maximum each bin has reached, decaying slowly
Area fill	Filled to the baseline instead of a bare line
Gradient	Colour by amplitude instead of one flat colour
Passband mask	A translucent band showing the receive passband, wired to the live filter edges
Centre / tuned / spectrum peak markers	Three optional markers

Levers: the whole DISPLAY tab, the SPECTRUM tab, the TRACE tab, and the `FILL`, `SG`, `SAR`, `ZOOM+/-` buttons.

Tap-to-tune. The spectrum and the waterfall behave as one tuning surface. Press, drag to position the cursor, release — the rig is retuned to the frequency under your finger, rounded to `TUNE SNAP`. The CAT set is fired **only on release**, never while dragging. If `FREQ TOUCH` is off, the cursor still tracks your finger (so the panel does not read as dead) but no command is sent.

During a voice over the spectrum strip and the waterfall area together become the **TX monitor**. It shows what actually goes to the uSDX, after the whole speech processor:



Figure 6.9 — TX monitor.

- **TX AUDIO** (centre panel): the spectrum of the transmitted audio, 0 to 3.5 kHz, labelled every 500 Hz, with level lines at −20 / −40 / −60 dBFS. The band from **300 Hz to TX HICUT** is shaded like the receive passband mask, with both edge values written at the top — speech should sit inside it.

- **Left bars: MIC**, the level after MIC GAIN (turns red above −3 dB); **OUT**, the level sent to the uSDX (turns red when the output limiter holds the ceiling).
- **Right bars: EXP** attenuation, **COMP** gain reduction, **CLIP** percentage — the MIC tab stages in chain order (§8.12). A stage that is switched off shows **OFF**.
- **Level history** (waterfall area): the last 15 seconds or so of the over, newest on the right, from +10 down to −60 dBFS. **OUT** is the filled green area and **MIC** the yellow line over it; the dashed red line is the output limiter's ceiling. A well-processed voice shows a jumpy MIC line over a dense, even OUT area that touches the ceiling without flattening against it. The strip underneath shows **GR**, the compressor's gain reduction (0 to 20 dB, bars hanging down), and red **CLIP** ticks wherever the clipper acted — so you can see exactly which syllables the compressor and the clipper are working on.

The bars hold each peak and fall back at about 25 dB per second. When the PTT is released the normal spectrum returns on the next frame, and the waterfall comes back with its history intact — it is only covered during the over, never overwritten.

6.10 Waterfall (270–405)

135 rows of history. Each bin is coloured against **its own tracked noise floor**, not against one global level — which is what lets a weak signal stay visible next to a strong one. Seven palettes, cycled with `WFG`.

Levers: `WATERFALL SPEED`, `WF BLACK LEVEL`, `WF CONTRAST`, `WF FLOOR BLEND` on the SPECTRUM tab; `WF FLOOR` per band on the CALIB tab.

> Changes to the waterfall are **not** visible while the menu is open — the menu covers that region and its repaint is suppressed. Close the menu and let the old history scroll off (about 3 s at zoom ×1, about 23 s at ×8) before judging a change.

6.11 Frequency scale

The absolute frequency under each part of the spectrum, computed from the CAT LO plus the offset, with the sideband sense taken into account.

6.12 CW decoder / RSS line (424–449)

One line of text. In CW it carries the decoded characters; otherwise it can run an RSS ticker, cycled with the `RSS` button.

Levers: the whole CW tab.

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7. The button rows

Two rows of ten. Buttons with an LED bar are toggles and the bar shows the state; the others are momentary and step through a list, with the label acting as the readout.

7.1 Top row — receive controls (y 127–157)

Button	Type	What it does
ATT	toggle	Input attenuator
AGC	toggle	Audio AGC. With it off the chain falls back to the plain `VOLUME` trim, which at its maximum of 1.0 is quiet — that is expected, not a fault. `VOLUME` is an attenuator only, so it cannot make up the difference.

Button	Type	What it does
NB	toggle	Wideband impulse noise blanker, applied to I/Q **before** decimation. Threshold on the AUDIO tab.
DNR	toggle	Adaptive (NLMS) noise reduction — keeps what is *periodic*. Good on hiss under a voice.
FDNR	toggle	Frequency-domain (spectral) noise reduction — estimates noise per FFT bin. A **different** tool from DNR, not a stronger one. Ignored in CW. Which of the three engines it runs is set by FDNR ALGO on the AUDIO tab; the default needs no attention.
NOTCH	toggle	Automatic notch — the same NLMS predictor as DNR, but keeping the *error* instead of the prediction. Kills a steady carrier.
FILTER	momentary	Steps the receive bandwidth. The list is per mode class: **SSB** 3600 / 2800 / 2400 / 1800 / 1200 Hz; **CW** 1000 / 500 / 250 / 100 / 50 Hz; **AM** 4500 / 3500 / 2500 Hz. Each class remembers its own selection. The label *is* the readout.
I/Q	momentary	Cycles the sample rate, and therefore the visible span: 8k → 32k → 48k → 96k. Re-clocks I²S, reprograms the WM8731, and is remembered across reboots.
FILL	toggle	Spectrum fill-to-baseline
MUTE	toggle	Mutes the output, gated after the AGC so unmuting does not arrive at full blast. Not persisted — a rig that boots muted reads as broken.

> **DNR and FDNR are switched separately on purpose.** The point of having both is to A/B them on the same signal, which is why they sit next to each other on the row.

7.2 Bottom row — display and system (y 450–480)

Button	Type	What it does
VOL+ / **VOL−**	momentary, repeat	Audio level, 0.05 … 1.00 in steps of 0.05. The same value as `VOLUME` on the AUDIO tab — an open menu shows the change immediately.
SSTV	toggle	Opens and closes the decoder window — SSTV, WeFax, RTTY, PSK, FT8 (§7.3). Its LED shows the window is open.
ZOOM+ / **ZOOM−**	momentary	Zoom-FFT factor, 1 to 8. Real narrow bins, not interpolation. The S-meter deliberately stays on the main spectrum.
WFG	momentary	Cycles the waterfall palette (7 entries)
SG	toggle	Spectrum gradient — colour by amplitude vs. one flat colour
SAR	toggle	Spectrum Auto dB Range — lets the display window track the signal automatically
RSS	momentary	Cycles the RSS ticker feed; LED off when the feed is OFF
MENU	momentary	Opens and closes the settings overlay

7.3 The decoder window (SSTV / WeFax / RTTY / PSK / FT8)

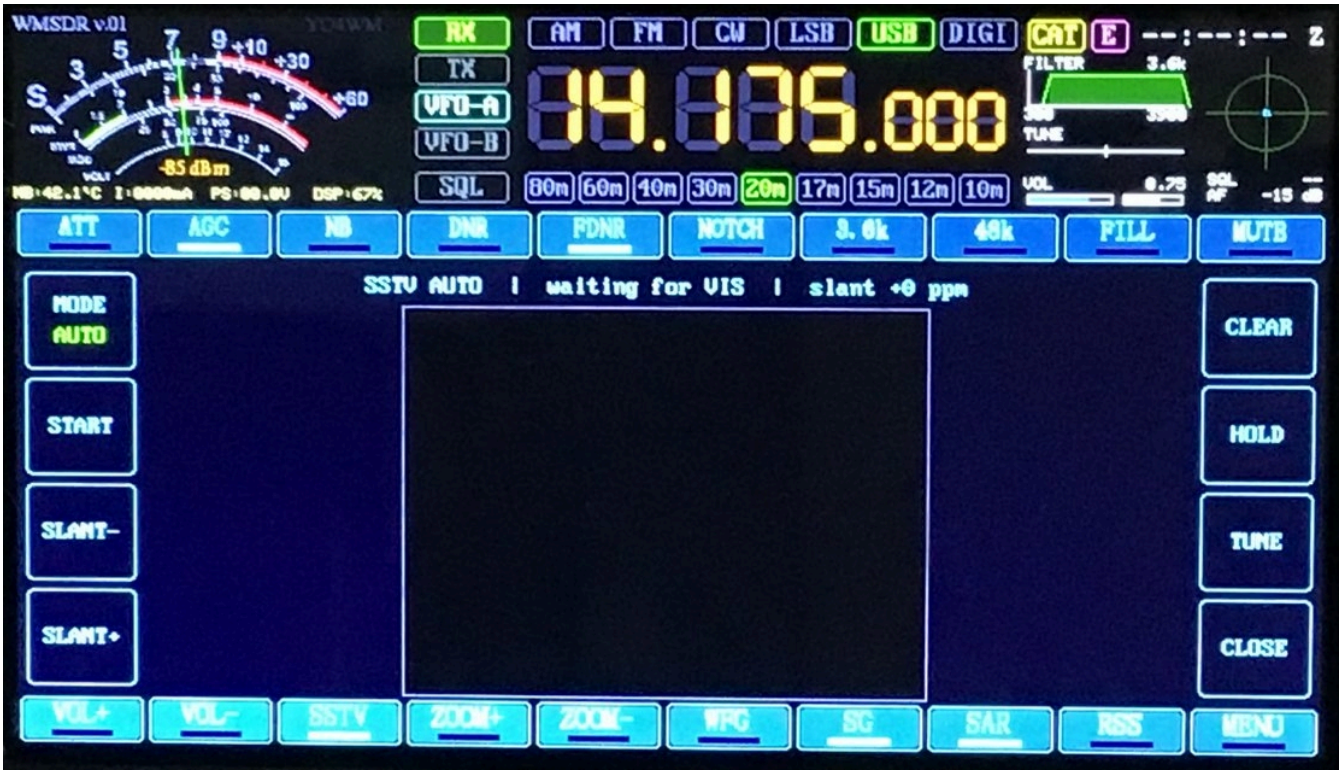


Figure 7.3 — The decoder window, MODE AUTO, waiting for an SSTV VIS header. Mode and actions on the left, CLEAR / HOLD / TUNE / CLOSE on the right.

A receive-and-display decoder for slow-scan TV pictures, HF weather charts, RTTY and PSK31/PSK63 text, and FT8. It opens
in the same place as the menu and replaces it — the two are never open together. Nothing is stored on the radio: the board has no SD card, so a picture lasts until it is cleared or scrolls away.
WeFax charts and FT8 decodes can be saved from a browser through the companion (§7.4).

> **RTTY, WeFax, PSK31 and FT8 are verified on the air. SSTV is not yet verified on a real transmission** — it passes its built-in self-test (see *Self-test* below).

Layout. Four buttons in a column on each side, the picture in the middle, a status line above it.

Left column	Right column
MODE — what to decode; the label shows the current choice	**CLEAR** — blank the picture and wait again
START — start without waiting for a header, or stop	**HOLD** — keep the picture; LED lit while on
SLANT− — trim the timing by −20 ppm	**TUNE** — tone bar instead of the picture; LED lit while on
SLANT+ — trim the timing by +20 ppm	**CLOSE** — close the window (the SSTV button does too)

MODE cycles through:

Setting	Decodes
AUTO	SSTV; the VIS header picks Martin 1 or Martin 2
M1 / **M2**	SSTV; the VIS header still decides — the setting is what START uses without one

Setting	Decodes
WEFAX	HF radiofax, 120 lines/min, IOC 576
TEST	self-test: plays a synthetic Martin 1 picture through the decoder
RTTY	RTTY text (Baudot); speed and shift from the preset
PSK	PSK31 / PSK63 text (Varicode); pick the signal on the audio waterfall
FT8	FT8, every 15 s UTC slot; a list of decodes

Span. SSTV needs 12 kHz audio. If the span is 8k or 32k, opening the window switches it to **48k** and closing puts it back. The switch is not saved — a reboot comes up on your own span. If you press I/Q while the window is open, your choice stands. At 96k nothing changes.

The audio is untouched. The decoders take a copy of the audio after the FILTER bandwidth and before NR, notch and AGC, so those settings do not affect decoding, and you keep listening normally.

TUNE — the tone bar

A bar from 1000 to 2500 Hz with markers at **1200** (sync, red), **1500** (black), **1900** (SSTV header, yellow) and **2300** (white). The green needle is the measured tone and the grey band its spread over the last half second, with the frequency and level underneath. It greys out when there is no tone.

- **SSTV** as it starts: the needle rests on **1900**, then darts between 1200 and 1500–2300.
- **A weather chart**: swings between **1500** and **2300**.
- **Voice**: wanders and the grey band breathes — expected, voice is not a single tone.

Receiving SSTV (Martin 1 / Martin 2)

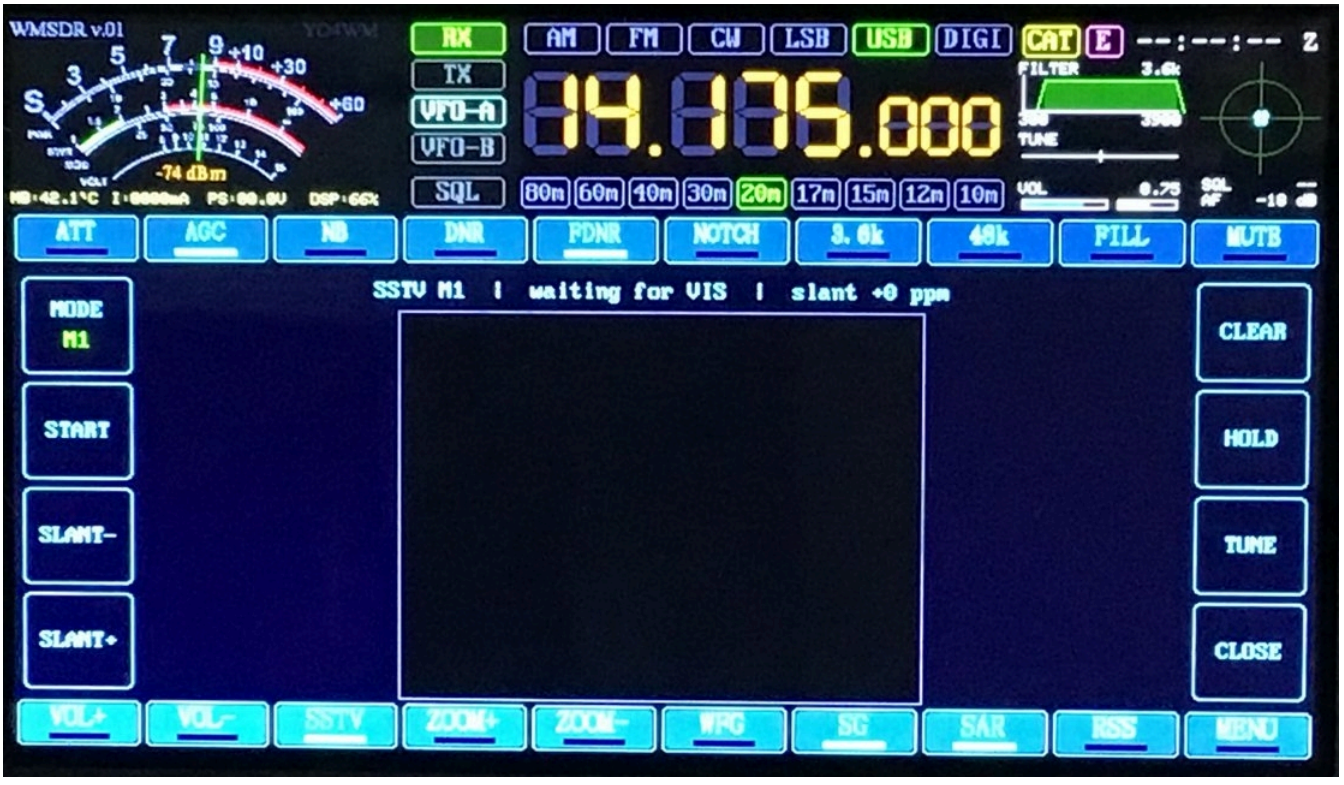


Figure 7.3a — MODE M1 (Martin 1) selected, waiting for a signal.

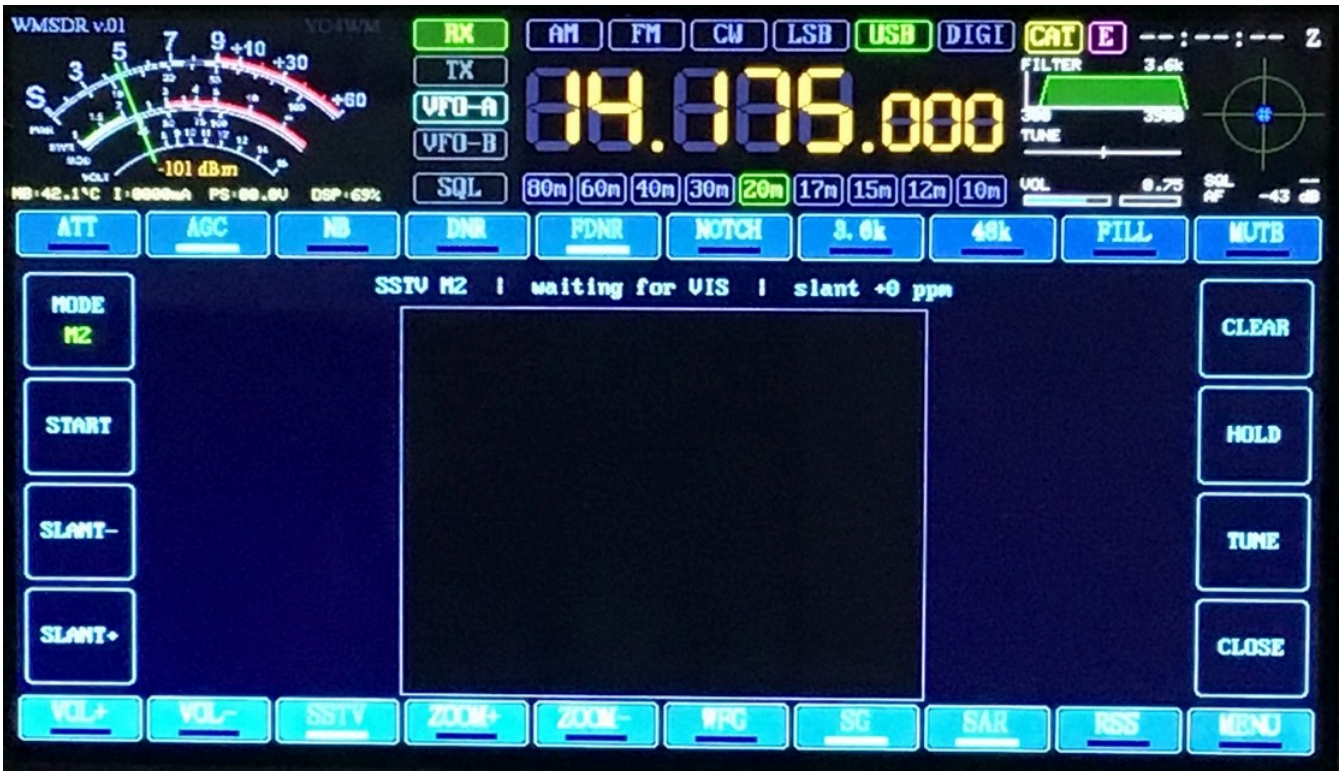


Figure 7.3b — MODE M2 (Martin 2) selected, waiting for a signal.

1. Tune the rig as for voice — USB on 20 m and up, LSB on 40 and 80 m.
2. Open the window, MODE **AUTO**, HOLD off.
3. When a picture starts, the header is decoded (`VIS 44 -> M1` in the serial log) and the picture builds from the top: **114 s** for Martin 1, **58 s** for Martin 2.
4. The status line shows `RX line n/256`, how many sync pulses were found, and the **slant** in ppm. The slant is corrected automatically: it moves over the first few dozen lines, then settles.
5. At the end: `complete`. The picture stays until the next header replaces it — press **HOLD** to keep it.

If the header is missed (weak or fading signal), press **START** as the picture begins; it finds the line sync on its own. If the image leans, **SLANT±** trims it. After 12 lines without sync the picture stops with `signal lost` and keeps what it has.

Receiving WeFax



Figure 7.3c — MODE WEFAX selected, waiting for a chart.

1. Tune **USB, 1.9 kHz below the published carrier** (see *Where to listen*). On TUNE, a chart swings between 1500 and 2300.
2. Open the window, MODE **WEFAX**.
3. A chart begins with a 5 s start signal (`start signal detected`), then 30 s of phasing (`phasing n/56`). Phasing measures the **clock error** and lines the chart up; the status line then shows `RX line n` and the clock in ppm.
4. The chart appears as a **strip scrolling upwards**, newest row at the bottom. Each row is three chart lines, which keeps the proportions. It ends on the stop signal (`chart complete`).

Tuned in mid-chart? Press **START**. Lines are received without phasing, so the chart comes out wrapped sideways: tap the strip where the chart's left margin appears, and that column becomes the left edge. Worth knowing about tapping:

- only rows from then on move — rows already on screen stay as they were;
- it shifts one way only: to move the margin a little to the right, tap near the right end;
- a fingertip is several columns wide — tap again if it lands slightly off;
- straighten any lean with **SLANT±** first, or the margin drifts away again;
- taps count only while a chart is being received, and any stray tap on the strip shifts it.

HOLD freezes the strip while reception continues; **CLEAR** blanks it and waits for the next start signal.

> **Verified example — DWD Pinneberg, 7880 kHz:** dial **7878.1 kHz USB**, MODE **WEFAX**. Charts received cleanly and saved from the browser (§7.4).

Receiving RTTY



Figure 7.3d — MODE RTTY selected, waiting for a signal.

Radioteletype text — amateur RTTY and weather bulletins — in an 80 × 16 console that scrolls upwards. In the RTTY modes three buttons change their job:

Button	In RTTY
REV (START)	swap mark and space; LED lit while on
TONE− / **TONE+** (SLANT−/+)	move the expected tone pair by 10 Hz
CLEAR	clear the text
HOLD	freeze the console; decoding continues
TUNE	tone bar with **M** (mark) and **S** (space) markers at the live tone pair

Tap the status line to step through the speed / shift presets:

Preset	Used by	Mark / space audio
45.45/170	amateur RTTY (default)	2125 / 2295 Hz
50/170	some utility and amateur stations	2125 / 2295 Hz
50/450	DWD weather bulletins	1675 / 2125 Hz
75/170	faster amateur / utility traffic	2125 / 2295 Hz
50/850	older wide-shift utility stations	1475 / 2325 Hz

The status line reads, for example, `RTTY 50/450 | M 1714 S 2164 | AFC +39 Hz | LTRS`.

1. Tune USB so the two tones sit near the **M** and **S** markers on TUNE — the AFC then pulls them in, up to ±200 Hz, and the status line shows how far (`AFC +n Hz`).
2. Pick the preset by tapping the status line.
3. Text appears as it is received; each finished line is also written to the serial log.

4. Nothing but garbage on a clean signal? Press **REV** — stations differ in which tone is mark, and changing sideband swaps them too.

Decoding details: letters/figures shifts are followed, a space returns to letters (the usual convention), and a character with a bad start or stop bit, or too weak, is dropped rather than printed as rubbish.

> **Verified example — DWD Pinneberg, 10100.8 kHz:** dial **10098.9–10099.1 kHz USB**, preset **50/450**,
> **REV off**, AFC settled at **+39 Hz**. Clean copy of the station's own bulletin, including its frequency list. An AFC offset of that size on every station points at the rig's frequency calibration; **TONE+** can pre-set it.

Receiving PSK31 / PSK63

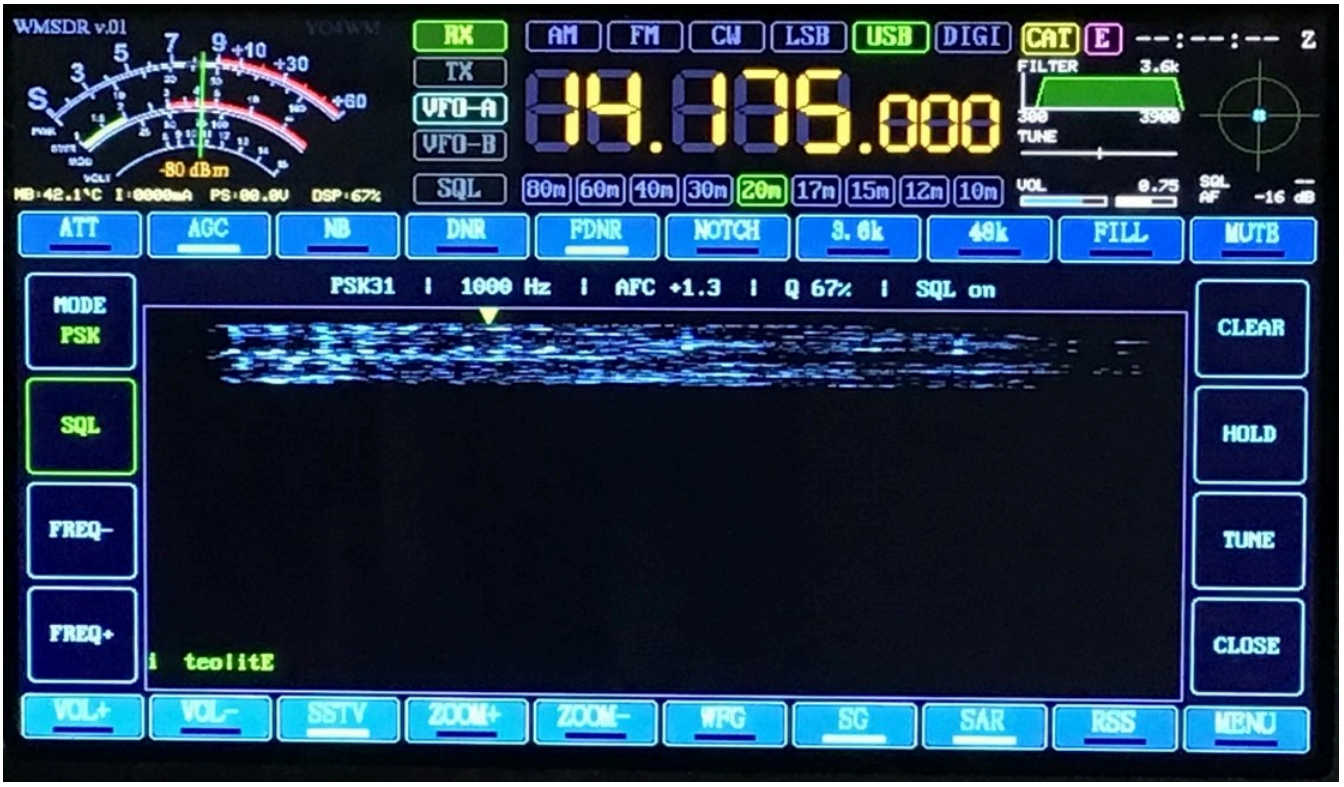


Figure 7.3e — MODE PSK selected, waiting for a signal.

Keyboard-to-keyboard PSK text. Several signals usually share the same 3 kHz of audio, so this mode shows an **audio waterfall** (0–3000 Hz) under the status line, a green marker above it for the signal being decoded, and the last 12 lines of text below.

Button / area	In PSK
SQL (START)	squelch; LED lit while on (default on) — drops characters when the signal quality is below 60 %
FREQ− / **FREQ+** (SLANT−/+)	move the decoded frequency by 5 Hz
CLEAR	clear the text
HOLD	freeze the text; decoding continues
TUNE	tone bar with a **PSK** marker at the decoded frequency
tap the **waterfall**	decode the signal under your finger
tap the **status line**	switch between **PSK31** and **PSK63**

The status line reads, for example, `PSK31 | 1000 Hz | AFC +2.4 | Q 92% | SQL on`.

1. Tune USB to the PSK activity — the traces appear as thin vertical lines on the waterfall.
2. Tap a trace. The AFC pulls in the last few hertz (up to ±30 Hz) while the quality is good.
3. **Q** is the signal quality: above ~60 % text prints; a clean signal reads 90 % or more.
4. Only noise with SQL off, or nothing with SQL on? The marker is probably off the trace — tap it again or trim with **FREQ±**. A wide, "double" trace is usually PSK63: tap the status line.

Each finished line of text is also written to the serial log.

Receiving FT8

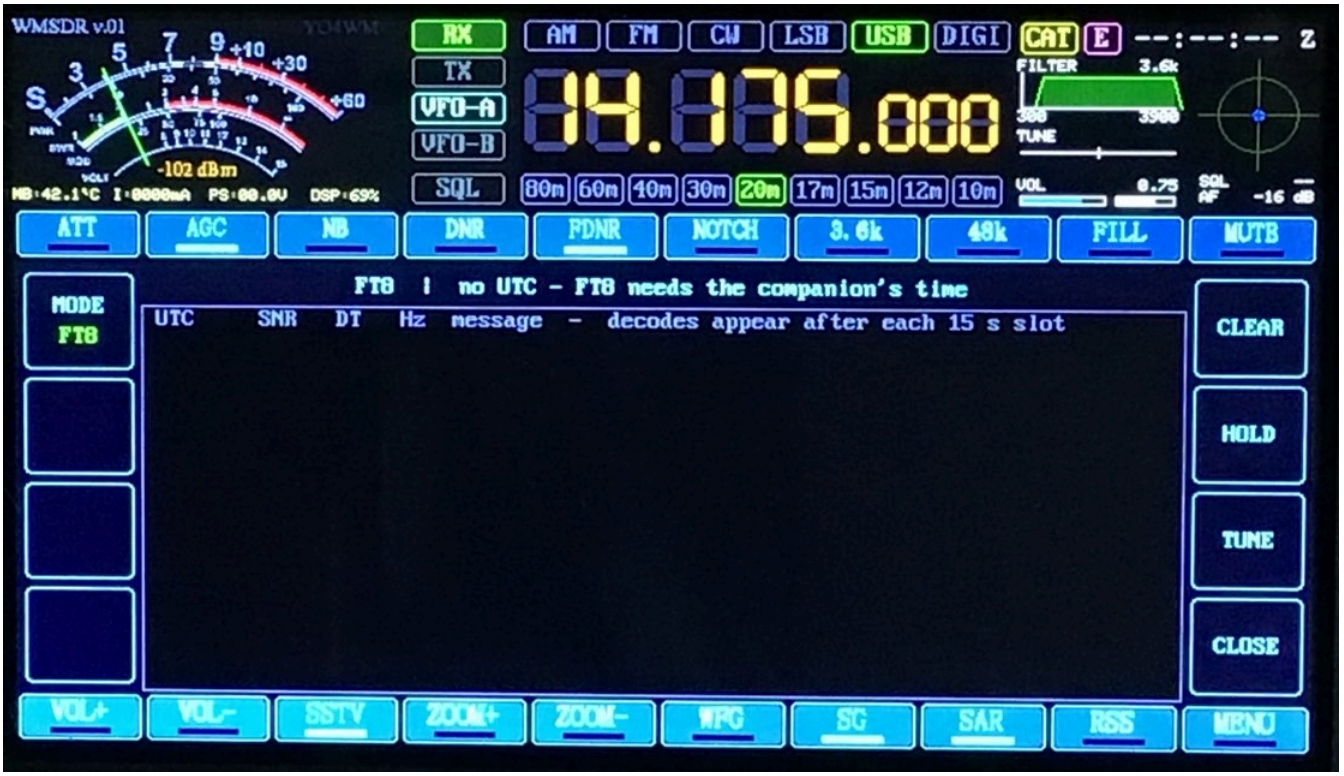


Figure 7.3f — MODE FT8 with no UTC yet: the header says so until the companion supplies the time. Decodes appear after each 15 s slot.

FT8 needs no tuning inside the window and no START: every transmission lasts 15 s and starts on a UTC boundary (:00, :15, :30, :45), so the radio records each slot and decodes it when it ends.

Two things it needs:

- **UTC from the companion** (the **E** icon, §6.5). FT8 slots must start within a fraction of a second, and the radio has no clock of its own. Without it the status line reads `no UTC - FT8 needs the companion's time` and nothing is decoded.
- **12 kHz audio** — arranged by the window itself (see *Span* above).

1. Tune **USB** to an FT8 frequency (see *Where to listen*), e.g. **14.074 MHz**.
2. Open the window, MODE **FT8**.
3. The status line counts through `waiting for slot hh:mm:ss`, then `slot hh:mm:ss` while it records, then `decoding`. Decodes appear a few seconds after each slot ends.

The list shows the last 16 decodes, newest at the bottom:

```
UTC     SNR   DT   Hz  message
111415   -5 +0.1  822  CQ ON3URT JO10
111415  +11 +0.1 1706  KH8WW IW4EGP JN64
```

- The first line of each slot is **white**, the rest green, so slots read as groups.
- **SNR** is the decoder's own estimate. It is not calibrated like WSJT-X's, so compare stations with each other rather than with another program.
- **DT** is the station's timing within the slot. The radio measures its own audio delay from the busy slots and corrects it, so after a minute or two most stations sit near **0.0**; a station at +1 or +2 s has a clock of its own that is off.
- **Hz** is the audio frequency — where the station sits in the passband.
- `<...>` is a callsign sent only as a short code. The radio remembers the full calls it has heard, so these fill in (e.g. `<YL3KZ>`) once the station has appeared in full.

Button	In FT8
START, SLANT−/+	not used (blank) — FT8 runs on its own UTC slots
CLEAR	clear the list
HOLD	freeze the list to read it; decoding continues

The status line also shows the previous slot, e.g. `last 23 in 7010 ms`. Every decode, and a summary per slot, is also written to the serial log.

What to expect. On a busy 20 m band, typically **15–27 decodes per slot**. The radio decodes fewer than WSJT-X on a PC: the weakest signals, which WSJT-X digs out with several passes, stay below what the ESP32-S3 can process in one slot.

> **Verified on the air — 14.074 MHz, September 2026:** 16–27 decodes per slot, every slot decoded, DT settled at about 0, hashed callsigns filling in within a few slots.

The decodes also appear live in a browser, where they can be filtered and saved (§7.4).

Where to listen

What	Frequency	Mode	Notes
SSTV	3.730–3.735 MHz	LSB	European evenings
SSTV	7.165 MHz	LSB	busy at weekends
SSTV	14.230 MHz (also 14.227, 14.233)	USB	the main worldwide frequency
SSTV	21.340 / 28.680 MHz	USB	when the band is open
WeFax, DWD Pinneberg	dial **3853.1** kHz (carrier 3855)	USB	typically at night
WeFax, DWD Pinneberg	dial **7878.1** kHz (carrier 7880)	USB	day or night
WeFax, DWD Pinneberg	dial **13880.6** kHz (carrier 13882.5)	USB	typically by day
RTTY, amateur	14.080–14.100 MHz	USB	45.45/170; try REV
RTTY, DWD Pinneberg	dial **4581.1 / 7644.1 / 10099.1** kHz (carriers 4583 / 7646 / 10100.8)	USB	50/450; confirmed on air on 10100.8
RTTY, weather	dial **11037.3** kHz	USB	50/450; confirmed on air
PSK31	14.070 MHz	USB	daytime
PSK31	7.040 / 3.580 MHz	USB	evenings
FT8	**14.074** MHz	USB	the busiest; confirmed on air
FT8	7.074 / 3.573 MHz	USB	evenings and night

What	Frequency	Mode	Notes
FT8	21.074 / 28.074 MHz	USB	when the band is open

Self-test: TEST

With MODE on **TEST**, **START** plays a perfect synthetic Martin 1 picture straight into the decoder — the receiver is not involved — with a deliberate +50 ppm clock error: header `VIS 44 -> M1`, then 8 vertical colour bars (white, yellow, cyan, green, magenta, red, blue, black); the slant settles near +50 ppm.

If it comes out right, the decoder is fine and any problem on the air is tuning or propagation. The self-test runs slower than real time; that does not change the result.

Limits

- Nothing is saved on the radio — WeFax charts and FT8 decodes can be saved from a browser (§7.4); SSTV pictures, RTTY and PSK text cannot yet.
- SSTV: Martin 1 and Martin 2 only. WeFax: 120 lines/min, IOC 576 only. RTTY: Baudot (ITA2) only — no SITOR / NAVTEX. PSK: BPSK31 and BPSK63 only — no QPSK. FT8: receive only — no FT4, no transmit.
- The TEST mode is temporary and will be removed once SSTV has been verified on real signals.

7.4 Companion web pages (beesdr-dx.local)

The radio has no storage and no keyboard. The companion XIAO — the board that already brings UTC and DX spots over ESP-NOW — receives the decoded WeFax charts and FT8 decodes, keeps the latest ones in its memory and serves them to any browser on your Wi-Fi; it also carries CW typed in the browser to the radio. Saving happens in the browser, on your phone or PC.

Open `http://beesdr-dx.local/`:

- **Home** is a dashboard with live tiles — UTC, FT8 (decodes in the last slot), Pictures, DX cluster, and the companion's Wi-Fi signal and uptime. Tap a tile to open its page.
- ☰ (top left, on every page) opens a drawer with all pages: Home, FT8, Pictures, **Transmit**, Control panel, RSS feeds, Settings, Status (JSON).
- ☀ / ☾ (top right) switches between a dark and a light theme; the choice is remembered in that browser.
- **Settings** holds the Wi-Fi, **CALLSIGN** and DX-cluster fields. Saving reboots the companion; leaving the password field empty keeps the stored one. In first-time setup (the `BEESDR-DX-Config` access point, 192.168.4.1) the settings form opens directly.

The live pages show **LIVE** while connected and reconnect by themselves.

Needed: the radio and the companion on current firmware, the companion on your Wi-Fi, and the decoder window open in the mode concerned — the companion only receives while the radio decodes.

Pictures — `/fax.html`

- The WeFax chart **builds live** as it is received, row by row; **follow live** keeps the newest row in view.
- The last **2 charts** are kept until the companion restarts; click one in the list to show it. A page opened mid-chart loads what has been received so far.

- **Save PNG** saves the chart shown, named by date and time (e.g. `wefax_20260917_1029Z_3.png`).
- A chart is 640 pixels wide, as on the radio's screen.

> **Verified example — DWD Pinneberg, 7880 kHz:** dial **7878.1 kHz USB**, MODE **WEFAX**. The "Atlantic North: sea state" chart was received live in the browser and saved as PNG.

FT8 — `/ft8.html`

Part	What it does
table	newest slot on top: UTC, SNR, DT, Hz, message
green message	a **CQ**
orange line	a message with **your call** — the **CALLSIGN** set on the companion's config page
CQ only / **my call only**	filters
search box	shows only messages containing a call or grid, e.g. `JN`
Pause	freezes the table to read it; decodes keep arriving, the button counts them, click again to jump to the latest
`last slot …: N decoded, M received`	the radio's count against what reached the page; **red** when they differ — decodes were lost on the link
Save log	downloads all stored decodes as a text file in WSJT-X `ALL.TXT` style

The companion keeps the last **1000 decodes** until it restarts; reloading the page brings them all back.

Transmit (CW) — `/tx.html`

Send CW from any browser — phone, tablet or PC. The radio keys the uSDX through its PTT line; the page only types.

The radio decides. Nothing is sent unless **TX ARM** is on at the radio (MENU → TX, §8.11). The badge at the top shows what the radio reports, once a second:

Badge	Meaning
RADIO NOT HEARD	no status from the radio for 4 s — link or radio down
NOT ARMED	the radio refuses text; arm it at the radio
ARMED (orange)	ready
SENDING (red)	keying, with the characters still queued

Live keying (on by default) sends every character **as you type it** — no Enter, so there are no gaps for another station to jump into.

- **Enter** types a space.
- **Backspace** takes back letters the radio **has not started** yet; letters already on air cannot be recalled.
- Only characters CW can send are accepted: A–Z, 0–9, `. , ? / = + - @`, and prosigns in angle brackets — `<AR> <SK> <BT> <KN> <AS> <HH>` (a prosign goes out once its `>` is typed).
- If you type slower than the keyer, the over **hangs 1.5 s** between words (sidetone on, receive muted) instead of ending after every word.
- It works from phone keyboards too.

With live keying off, type a whole line and press **Enter** or **Send**.

- **Speed** — the WPM slider (5–60) follows the radio and changes it when released.
- **Macros** — CQ, QRZ?, RST, 73 `<SK>`, MY CALL, AGN?; `{CALL}` is replaced by the CALLSIGN from Settings. *edit macros* changes them (`Label | text`, one per line, saved in that browser).
- **STOP** (or **Esc**) stops at once and clears whatever is still queued.
- **Sent** logs what went out, and in red anything refused because the radio was not armed.

> **Verified on the air (September 2026):** live typing to 45 WPM, clean on the sidetone and on a second receiver.

If a page stays empty

- **LIVE but nothing appears:** is the decoder window open in WEFAX / FT8 on the radio?
- **FT8 shows `N decoded, M received` in red often:** the ESP-NOW link is losing frames — move the companion closer or improve its antenna.
- **The companion cannot join the Wi-Fi at all** — it sees networks in a scan but connects to none, not even its own `BEESDR-DX-Config` access point: check its **antenna** first (the small snap-on connector comes loose easily), then its USB power. This happened on the prototype; a better antenna cured it.
- **Transmit shows RADIO NOT HEARD** after the router changed Wi-Fi channel: current companion firmware follows the channel by itself; older firmware needed a companion reboot. Fixing the router on one channel (11 is what BEESDR listens on first) avoids the search altogether.

7.5 Zeus / HPSDR mode

In Zeus mode the BEESDR stops being a panadapter and becomes a **network radio**. It shows up on your Wi-Fi as an **OpenHPSDR radio** (Protocol 1, the protocol of the Hermes and Hermes-Lite 2 boards) and streams the rig's I/Q to a program on a PC. The PC draws the spectrum and waterfall, demodulates and runs the station. The BEESDR carries the I/Q to it, plays its audio, keys the rig and keeps the rig on the program's frequency.

Tested programs (October 2026): Zeus, **Thetis** and **piHPSDR**. All three find the BEESDR on the network and use it as a radio. Other OpenHPSDR Protocol-1 programs should work but have not been tried.

Entering and leaving

- MENU → **DSP** → **ZEUS** → **ON**. The BEESDR saves the choice and **reboots at once** into Zeus mode.
- The choice **survives a power cycle**: a station set up for Zeus comes back up in Zeus.
- To return to the normal radio, **hold EXIT** (bottom right of the Zeus screen) for about **1 second**. PTT is released, the choice is cleared and the BEESDR reboots.

The Zeus screen

Text only: no spectrum, no waterfall, no buttons. The only touch control is EXIT.

Line	Meaning
`IP … - waiting for client` / `STREAMING`	The BEESDR's address on your Wi-Fi, and whether a program is connected
`EP6 …/s  err …`	I/Q packets per second sent to the PC (about **381/s** at 48 kHz), and send errors
`EP2 …/s  gaps …/s`	Packets per second received from the PC (also about 381/s), and how many went missing. A missing packet is covered by a short fade, not a click

Line	Meaning
`Zeus …  rig … Hz` + mode	The program's RX1 frequency, the frequency last sent to the rig, and the rig's mode

Settings in the program

- **Sample rate 48 kHz.** Zeus starts at 192 kHz; the BEESDR streams 48 kHz only.
- The BEESDR answers as a **Hermes** board, not a Hermes-Lite 2: an HL2 has no codec, so some programs send it no audio. In Zeus, set **Settings → Radio → Audio output = "Radio Speaker/headphone"**, otherwise no audio reaches the BEESDR.
- The PC and the BEESDR must be on the same network. The program finds the BEESDR by itself (discovery on UDP port 1024).

> **Wi-Fi network.** In Zeus mode the BEESDR joins your home Wi-Fi as a client. The network name and password are set **in the firmware build** (a file kept out of the source repository), not from the menu.

How each part works

- **Receive.** The rig's I/Q goes to the PC unchanged, and the PC demodulates. The PC's audio comes back over Wi-Fi and plays on the BEESDR's own output (speaker or headphones), so you still listen at the radio. Expect about **¼ second** of delay between the band and your ear.
- **Tuning.** The rig **follows the program**: tune in the program and the rig is retuned over CAT (the frequency is sent again every second until the rig reports it). It does **not** work the other way: the rig's own knob does not move the program, because Protocol 1 has no way to tell the PC.
- **Mode.** Protocol 1 does not carry the mode either. Set USB / LSB / CW **on the rig itself** to match the program.
- **TX ARM is automatic** in Zeus mode: armed while a program is streaming, disarmed as soon as it stops or goes silent for 3 seconds.
- **Voice.** The program's **MOX**, or the hand-mic PTT, keys the BEESDR's own voice transmit chain: the microphone on the BEESDR, the MIC and TX tab settings, the rig keyed on P0_0. It is exactly a normal voice over (§8.11). The PC's microphone is not used. MOX is ignored while the rig is in CW, so it can never key a carrier.
- **CW.** Comes from the **paddle** on the BEESDR (§8.11), not from the PC. With the rig in CW, the paddle keys the rig. The program's own CW keyer and macros are not used.
- **Not available in Zeus mode:** the BEESDR's own spectrum, decoders and knobs, and ESP-NOW, so the companion web pages (§7.4) and the browser Transmit page do not work either.

---

8. The menu, tab by tab

Press **MENU** on the bottom row. The overlay covers the spectrum, waterfall, frequency scale and decoder line — the button rows stay live, so **MENU** closes it again.

Thirteen tabs: DISPLAY · SPECTRUM · TRACE · DSP · AUDIO · EQ · CW · CALIB · COLORS · KNOBS · TX · MIC · STATUS

How the controls work

- **Cells** with an LED bar are on/off. One tap toggles.

- **Numeric rows** have `-` and `+` buttons with the value between them. The buttons grey out at the end of travel. **Press and hold** to repeat: one step immediately, then repeats after 400 ms, then faster after 1.5 s.
- The **action button** at the bottom centre changes meaning per tab — usually **RESET DEFAULTS**.

How settings are saved

Everything except the STATUS tab is written to flash as **one versioned record** (the TX and MIC tabs keep their own record, and TX ARM and TX TONE are never saved), committed when you **close** the menu — six changes are one write, and no change is no write at all. Settings changed outside the menu (the `WFG`, `SG`, `SAR`, `FILL`, `VOL` buttons, the SQL tap) are caught by a deferred autosave that fires 3 seconds after the last change.

> **RESET DEFAULTS is load-bearing, not a convenience.** Several of these values are persisted *and* capable of making the display unreadable — a spectrum floor that pushes the trace off the window, a text colour set to its own background. Once stored, a power cycle no longer rescues you. The reset button is the way back, and the min/max limits on each row are the other half of that guard.

8.1 DISPLAY tab



Figure 8.1 — DISPLAY tab.

A 3 × 4 grid of on/off cells plus one numeric control. Everything here is about **what is drawn on the spectrum**, not about the DSP.

Cell	What it controls
CENTER PK	Marker on the centre (LO) frequency

Cell	What it controls
TUNED PK	Marker on the tuned frequency
SPEC PK	Marker on the strongest peak in the span
NOISE FL	Draw the tracked noise-floor layer
TRAIL	Draw the decaying trail layer
PEAK HOLD	Draw the peak-hold layer
LINE NF	Noise floor as a line (on) or filled (off)
LINE TRL	Trail as a line or filled
LINE PKH	Peak hold as a line or filled
FREQ TOUCH	**May a tap on the panadapter retune the rig?** With this off, the cursor still tracks your finger but no CAT set is sent. Turn it off if you are afraid of a stray touch during a contest.

Control	Range	Notes
BRIGHTNESS	20 – 255, step 5, shown as %	Panel backlight. The minimum is deliberately **not** 0: a backlight you can take to black would hide the very menu you need to bring it back.

Action: RESET DEFAULTS.

8.2 SPECTRUM tab

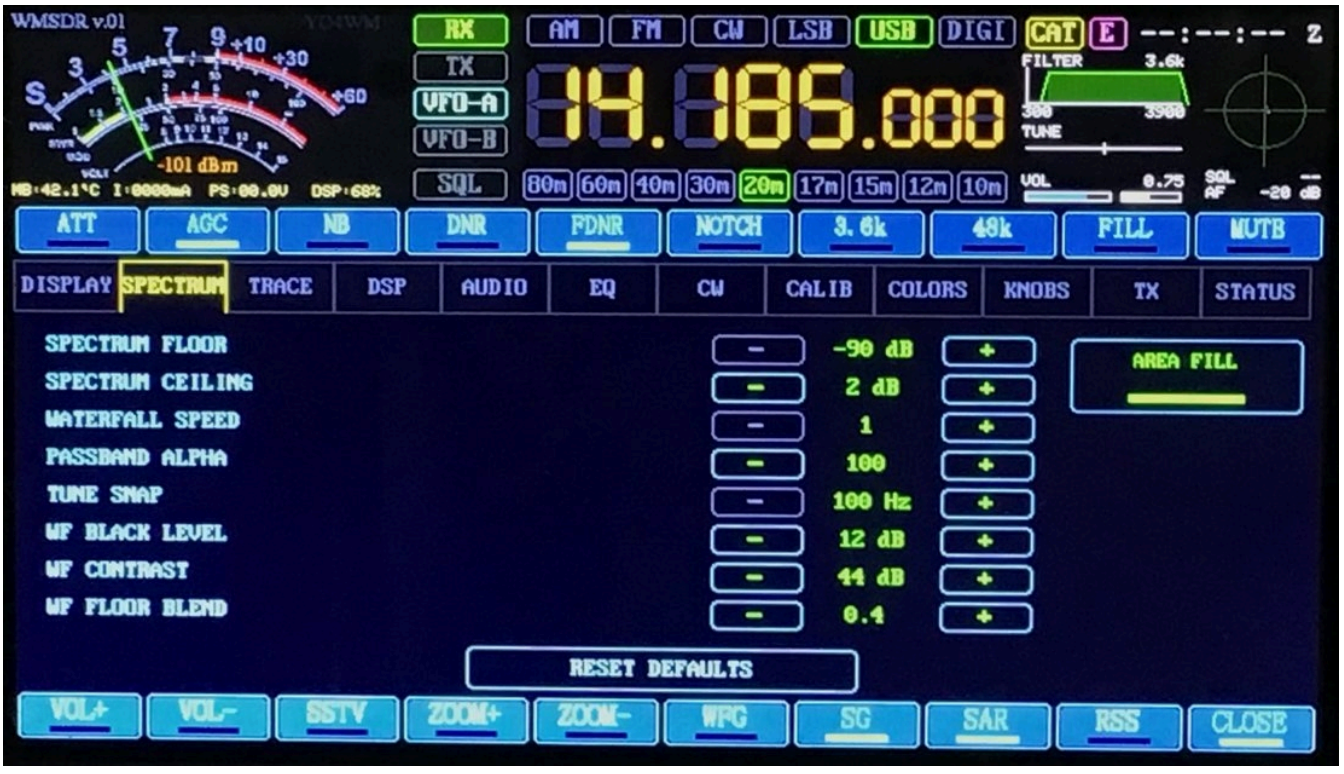


Figure 8.2 — SPECTRUM tab.

The display window and the waterfall's colour mapping.

Row	Range	Unit	What it does
SPECTRUM FLOOR	−90 … −20, step 1	dB	Lower clamp on the auto-ranging display window. Signals below this are at the bottom of the screen.

Row	Range	Unit	What it does
SPECTRUM CEILING	−20 … +40, step 1	dB	Upper clamp. Together with FLOOR, these bound how far the automatic range is allowed to travel — they are **not** the live window, which is recomputed every frame.
WATERFALL SPEED	1 … 5, step 1	—	Scroll rate. 1 is slowest and shows the longest history.
PASSBAND ALPHA	0 … 255, step 4	—	Opacity of the translucent passband mask over the spectrum. 0 makes it invisible — if you cannot see your passband, look here first.
TUNE SNAP	100 … 1000, step 100	Hz	Rounding applied to tap-to-tune. 100 for casual work, 500 or 1000 to land cleanly on a net frequency.
WF BLACK LEVEL	0 … 30, step 1	dB	How far above each bin's own noise floor the waterfall starts to show colour. Raise it to darken a noisy band.
WF CONTRAST	10 … 80, step 2	dB	The dB span that the palette is stretched across. Small = punchy and easily saturated; large = gentle.
WF FLOOR BLEND	0.0 … 1.0, step 0.1	—	How much of the **per-bin** floor is used versus one global floor. **0 collapses the scheme to a single global floor** — try that first if the waterfall ever looks wrong.

Cell	What it does
AREA FILL	Fills the area under the spectrum trace

Action: RESET DEFAULTS.

8.3 TRACE tab

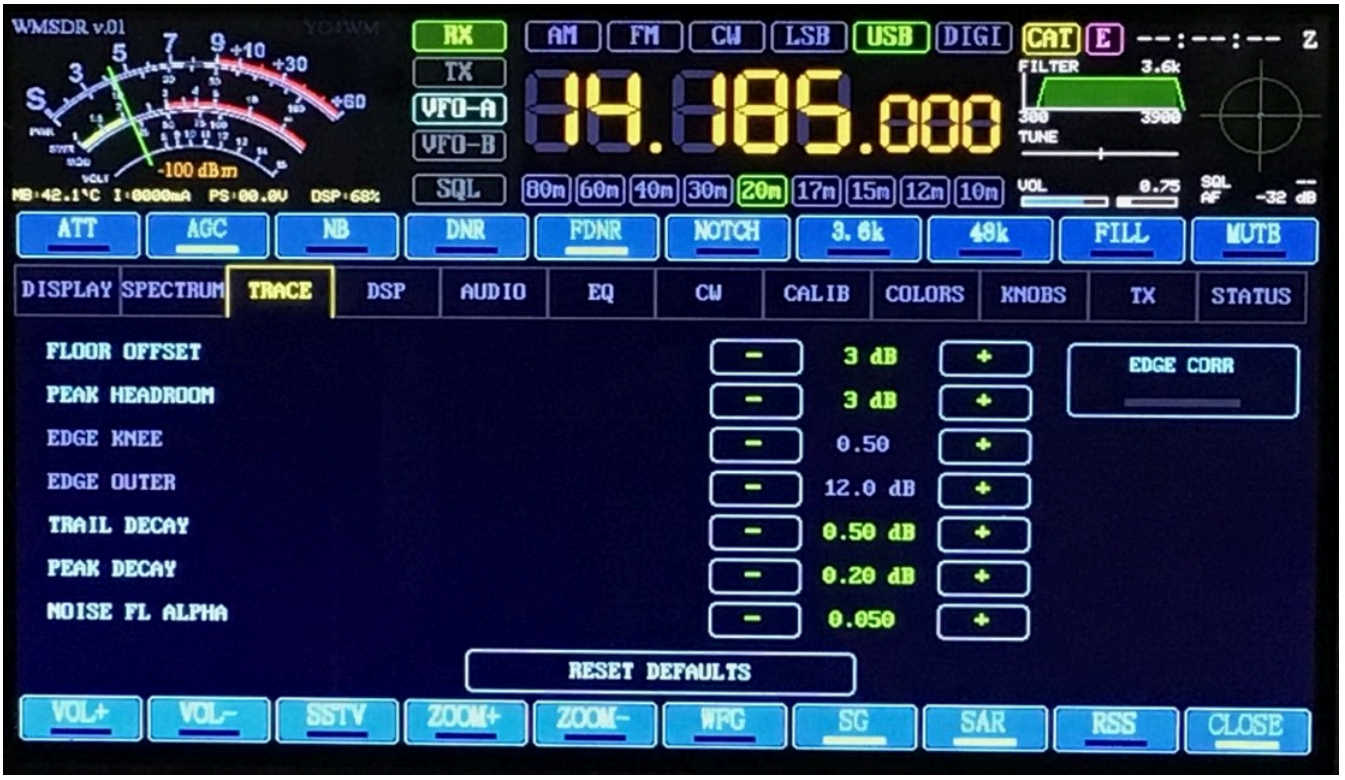


Figure 8.3 — TRACE tab.

The shape of the trace itself. These are tuned on the air against real band conditions, which is why they are persisted — and why the reset button matters here.

Row	Range	Unit	What it does
FLOOR OFFSET	−30 … +40, step 1	dB	Shifts the whole trace vertically. **The one to be careful with**: a large value pushes the trace clean off the visible window, and once stored a power cycle no longer clears it.
PEAK HEADROOM	0 … 30, step 1	dB	How much room is left above the strongest peak.
EDGE KNEE	0.10 … 1.00, step 0.05	—	Where the edge correction starts, as a fraction of half-span. *Greyed out unless EDGE CORR is on.*
EDGE OUTER	0.0 … 30.0, step 0.5	dB	How much correction is applied at the extreme edges. *Greyed out unless EDGE CORR is on.*
TRAIL DECAY	0.05 … 3.00, step 0.05	dB / frame	How fast the trail fades. Small = long tail.
PEAK DECAY	0.01 … 3.00, step 0.05	dB / frame	How fast peak hold falls back. Very small values effectively make it a permanent maximum.
NOISE FL ALPHA	0.002 … 0.200, step 0.005	—	How fast the tracked noise floor follows the band. Small = slow and stable; large = follows QSB and can swallow a weak steady carrier.

Cell	What it does
EDGE CORR	Enables the edge correction. The two EDGE rows are only live while this is on — greyed out otherwise, so they do not read as working controls that silently do nothing.

Action: RESET DEFAULTS.

8.4 DSP tab



Figure 8.4 — DSP tab. Photographed before the IQ BALANCE and ZEUS rows were added.

The spectrum smoothing chain, ahead of everything the other display tabs adjust — plus the CAT baud rate, which had to live somewhere reachable without a rebuild.

Row	Range	What it does
SPECTRUM EMA @48k	0.004 … 0.300, step 0.002	Exponential smoothing on the spectrum magnitudes, expressed at 48 kHz. Small = smooth and slow; large = twitchy and responsive. The tab prints the **effective alpha at the current sample rate** underneath, because the value scales with the rate and you should not have to do that arithmetic.
BIN AVG RADIUS	0 … 4, step 1	Averages each bin with its neighbours. 0 = off. Trades frequency resolution for a smoother-looking trace.
FFT WINDOW	list	The **display** FFT window: BLACKMAN-H · NUTTALL · BLACK-NUT · HANN · HAMMING · NONE. Listed low-sidelobe first, so moving down the list trades dynamic range for resolution. **Levels do not move between choices** — every window is gain-normalised to the Blackman-Harris reference, so the S-meter calibration stays valid whichever you pick. This is display-only; the receive chain has no FFT.
CAT BAUD	9600 / 38400 / 115200	The CAT serial rate. Applied to the live UART immediately, and restored before the port is opened at boot. Set it to match the rig.
IQ BALANCE	OFF / AUTO / MAN	Corrects the gain and phase mismatch between the I and Q channels, which otherwise leaks a mirror copy of every signal to the other side of centre. **AUTO** (default) measures it continuously from whatever is on the band and settles in a few seconds; it is frozen while you transmit. **MAN** applies the two rows below as set. **OFF** applies nothing. Corrects the spectrum and the audio alike.
IQ GAIN	−3.00 … +3.00 dB, step 0.05	The **board's** gain imbalance (Q against I), not the correction. Live in AUTO — the tab refreshes on its own. Stepping it while in AUTO switches to MAN, starting from the measured value.
IQ PHASE	−10.0 … +10.0 deg, step 0.1	The board's phase error away from 90°. Same behaviour as IQ GAIN.
ZEUS	OFF / ON	**ON reboots at once into Zeus mode** (§7.5): the BEESDR becomes an OpenHPSDR radio for a PC program. Stays in Zeus mode across power cycles until EXIT is held on the Zeus screen. Always reads OFF in the normal radio.

Reading the IQ rows. A well-matched front end shows a few tenths of a dB and under a degree — about 40 dB of mirror rejection before any correction. To null a mirror by hand: put a steady carrier a few kHz off centre, switch to MAN, step IQ PHASE for the smallest mirror, then IQ GAIN. The mode and both values are kept across power cycles, so AUTO starts from the last known balance instead of zero.

Action: RESET DEFAULTS (also returns IQ BALANCE to AUTO, 0 dB, 0°).

8.5 AUDIO tab



Figure 8.5 — AUDIO tab.

The receive audio chain — the part you actually listen to.

Row	Range	Unit	What it does
NB THRESHOLD	2.0 … 10.0, step 0.5	—	The impulse noise blanker's trigger point. **The most consequential setting on this tab**, because both failure modes are quiet: too high and the NB button does nothing at all; too low and it shaves the tops off voice peaks and CW keying, which sounds like mild roughness rather than a malfunction. The right value depends on the noise at your QTH.
AGC TARGET	0.05 … 0.25, step 0.05	—	The *average* output level the AGC aims for; default 0.20. Speech peaks run about three times higher, which is why the range stops at 0.25 — see the headroom note below.
AGC HANG	0 … 1000, step 50	ms	How long the gain holds after a peak before it starts to recover. Longer suits SSB voice; shorter suits CW.
AGC RELEASE	100 … 2000, step 100	ms	How fast the gain recovers after the hang expires.
AGC MAX GAIN	0 … +45, step 3	dB	Ceiling on the AGC gain; default **+34 dB**. In practice it decides **how loud the band noise gets in the pauses** compared with voice: at +34 dB the pause noise sits about 12 dB below the voice, at +45 dB it comes up nearly to speech level. Every step is audible; above about +42 dB the ceiling stops mattering on typical band noise.
DNR LEVEL	0.005 … 0.200, step 0.005	—	NLMS step size for the adaptive noise reducer. Higher tracks faster but roughens the audio; on voice, **"watery"** is the usual sign to back it off.
NOTCH LEVEL	0.005 … 0.200, step 0.005	—	NLMS step size for the automatic notch. Higher grabs a carrier faster but is more likely to chew on speech.
FDNR STRENGTH	0.00 … 1.00, step 0.05	—	How aggressively the spectral NR suppresses each bin.

Row	Range	Unit	What it does
FDNR FLOOR	−24 … −3, step 1	dB	How far down a bin is allowed to go. **If FDNR sounds like burbling rather than like quiet noise, move this UP (towards −3), not down.** Silencing bins completely is what creates musical noise; leaving an audible bed is what makes the residual sound natural.
FDNR ALGO	MIN-STAT / ROMANIN / **HYBRID**	—	Which noise-reduction engine the FDNR button switches on. **HYBRID is the default and there is no reason to move off it** — the other two are kept as reference points, so you can hear for yourself what each half of the algorithm contributes. See §8.5.1. Not saved: it returns to HYBRID at every power-up.
FM SQUELCH	0 … 100, step 1	—	FM only, ignored everywhere else. **0 = open** (the squelch does nothing), 1 = loosest, 100 = tightest. It gates on the hiss *above* the audio band, so the number is a **noise tolerance**, not a signal level — turn it up until an empty channel stays quiet, and stop there. Measured reference: about 0.17 on an empty channel, about 0.008 when fully quieted.
VOLUME	0.05 … 1.00, step 0.05	—	Output level after the AGC — an attenuator only, it never adds gain. The same value as VOL+ / VOL− on the bottom row.

> **Headroom: keep AGC TARGET × VOLUME at or below 0.25.** The AGC holds the *average* level; speech peaks run about three times above it, and above a product of about 0.28 they reach the output ceiling. That is heard as distortion that sounds like **reverb or echo**, worst in the pauses — and no AGC, noise or filter setting fixes it. The ranges above make it impossible to cross (0.25 × 1.00), and settings saved by older firmware are pulled into range at power-up. The AF readout (§6.8) shows where you are: about −6 dB on voice is healthy, −2 … 0 dB is too hot.
>
> **The FILTER bandwidth is deliberately not on this tab.** It is a per-mode list already shown as the FILTER button's own label; a numeric row here could only offer a bare index, which is worse than what the button already does.
>
> **FDNR on/off is the FDNR button**, not a row — it is a per-QSO control, and rows are expensive on this tab because every row's height is derived from the count.

Action: RESET DEFAULTS.

8.5.1 The three FDNR engines

Spectral noise reduction makes two decisions that are usually bundled together but are in fact independent: **how it estimates the noise**, and **how it treats the resulting per-bin gains**. BEESDR lets you hear each one separately, because the combination that works best is not the one either source algorithm arrived at on its own.

Choice	Noise estimate	Gain treatment	How it sounds
MIN-STAT	Running minimum of each bin's power over about 1.5 s. Signals come and go; the noise is the part that is always there, so the minimum is the noise.	Smoothed over time only.	Suppresses hardest — voice comes out clearest, residual noise lowest. But in the gaps between words it burbles: **a faint "radio station playing in the background".**
ROMANIN	Per bin, the probability that speech is present; the noise estimate moves only by the probability that it is *absent*.	Averaged across **neighbouring bins**, by an amount that widens the harder the filter is cutting.	No burbling at all, but the background noise is left noticeably louder.

Choice	Noise estimate	Gain treatment	How it sounds
	Reacts in tens of milliseconds instead of a second and a half.		
HYBRID *(default)*	MIN-STAT's.	ROMANIN's.	MIN-STAT's depth of cut with none of its musical noise: the noise drops and stays down, whether or not anyone is talking.

Why the burbling happens, and why HYBRID cures it: noise in a single FFT bin fluctuates by many decibels from frame to frame even when the noise itself is perfectly steady. Set a threshold and most bins fall below it while the occasional one spikes past and survives at full gain. **Those isolated survivors ringing on their own are what musical noise *is*.** Averaging each bin's gain across its neighbours cannot leave one standing alone. Smoothing in time — which is all MIN-STAT does — cannot fix it, because a bin that survives in every frame is perfectly steady in time and still a whistle.

The averaging is only applied when the filter is actually cutting hard, so audio on a quiet band is never needlessly blurred.

> **Ignored in CW**, all three of them. FDNR adds about 21 ms of delay and softens the edges of keying — which is exactly what the CW decoder measures marks and spaces from.

8.6 EQ tab



Figure 8.6 — EQ tab.

Four peaking bands on the demodulated audio.

Row	Range	What it is for
400 Hz	−12 … +12 dB, step 1	Bass / warmth. Cut it to fight rumble and LF noise.

Row	Range	What it is for
900 Hz	−12 … +12 dB, step 1	Body of the voice.
1600 Hz	−12 … +12 dB, step 1	Presence. Lift it for intelligibility on a weak signal.
2400 Hz	−12 … +12 dB, step 1	Top end. Lift for crispness, cut to tame hiss.

The centres are fixed at 400 / 900 / 1600 / 2400 Hz — chosen to sit *inside* a communications passband rather than on the ISO octave grid, so all four actually do something on SSB.

At 0 dB a peaking section is an exact identity, so **a flat EQ is bit-transparent and is skipped entirely** — leaving it flat costs nothing.

Action: RESET DEFAULTS.

8.7 CW tab

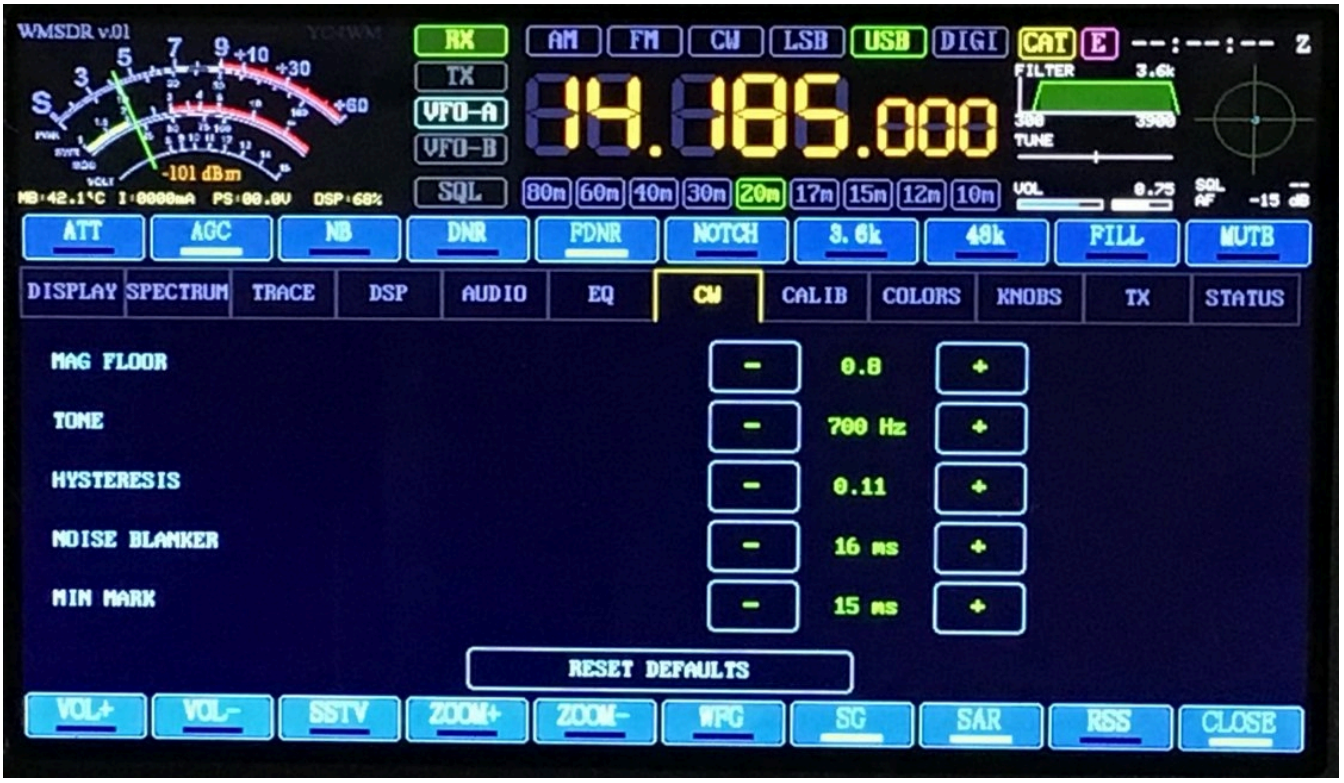


Figure 8.7 — CW tab.

Decoder tuning. The decoder text line on the main screen is your feedback, so every one of these can be judged on the air.

Row	Range	Unit	What it does
MAG FLOOR	0.1 … 20.0, step 0.1	—	The absolute magnitude below which the detector calls it silence. Too low and noise decodes as random characters; too high and weak signals are ignored.
TONE	400 … 1000, step 10	Hz	**The CW pitch for the whole radio**, not just the decoder. It re-centres the CW bandpass filter, moves the tuning aid, moves the spectrum marker, and sets the VFO offset — all from this one number.
HYSTERESIS	0.02 … 0.45, step 0.01	—	The symmetric gap between the on and off thresholds. Larger = more immune to noise flicker, but clips very fast keying.

Row	Range	Unit	What it does
NOISE BLANKER	0 … 50, step 1	ms	Ignores mark/space events shorter than this — impulse noise, not keying.
MIN MARK	10 … 120, step 5	ms	The shortest event accepted as a real dit. Sets the practical upper WPM limit.

The BPF bandwidth is **not** a row here: the decoder's pre-filter follows the **FILTER** button, so a separate control could only ever contradict it. Pick the CW bandwidth on the front panel and the decoder tracks it.

The defaults are the values calibrated on the air; RESET DEFAULTS returns to those.

Action: RESET DEFAULTS.

8.8 CALIB tab

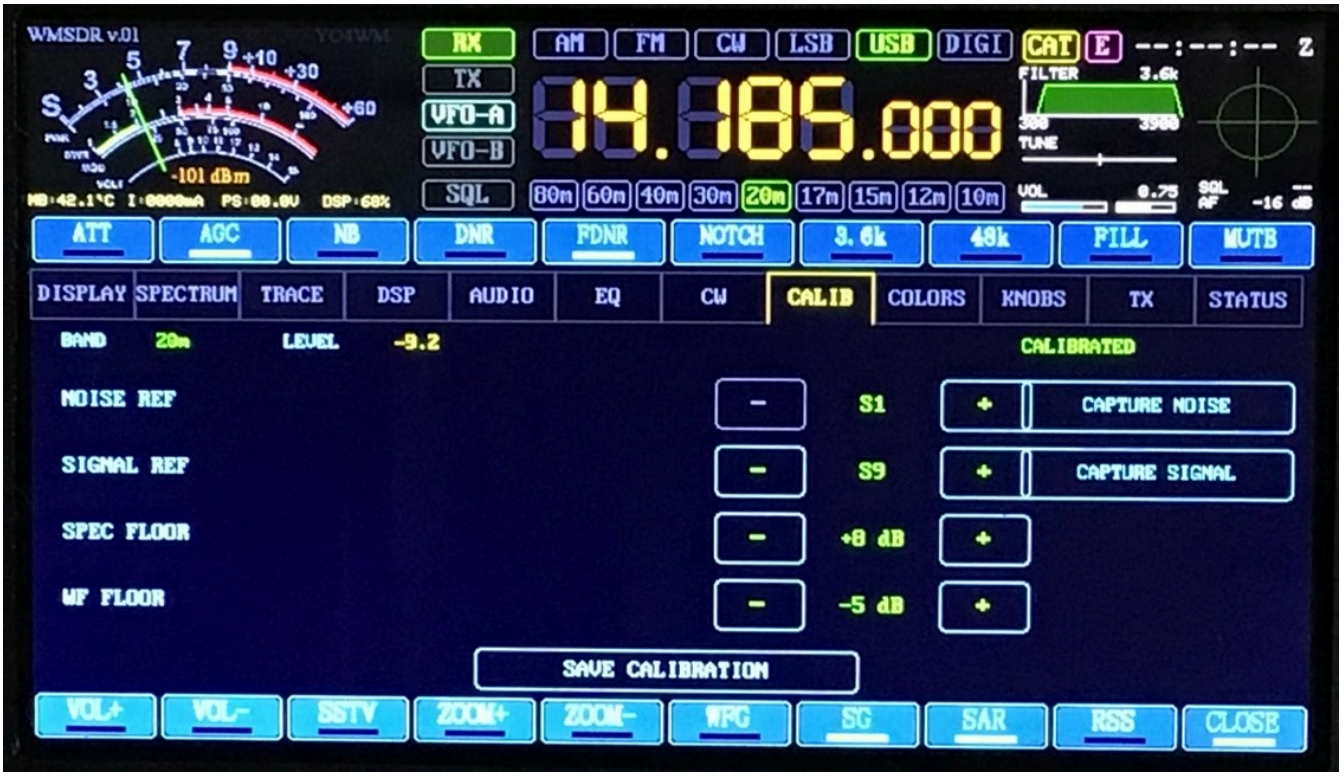


Figure 8.8 — CALIB tab.

Per-band S-meter and display-floor calibration. Flat controls, no wizard — every field and both capture buttons are always live, so you can redo a single point without walking a whole flow.

The header line shows the current **BAND**, the **live LEVEL** being measured (refreshed four times a second), and whether this band reads **CALIBRATED** or **UNCALIBRATED**.

Row	Range	What it does
NOISE REF	S1 … S15	The S-number you want the *quiet* reference point to read as.
SIGNAL REF	S1 … S15	The S-number you want the *strong* reference point to read as.
SPEC FLOOR	−30 … +30 dB	Per-band spectrum floor offset — lines the trace up across bands with different noise levels.

Row	Range	What it does
WF FLOOR	−30 … +30 dB	Per-band waterfall floor offset, same idea.

Two capture buttons sit on the two reference rows:

- **CAPTURE NOISE** — point the rig at a quiet part of the band, set NOISE REF (typically S1), press. The current level is stored as that S-number.
- **CAPTURE SIGNAL** — find a signal of known strength (or use a calibrated generator), set SIGNAL REF (typically S9), press.

Two points define the whole scale. Above S9 the scale continues at S9+10 dB per unit.

Action: SAVE CALIBRATION. The floor offsets are edited in place, so saving is a deliberate step — press-and-hold on a row would otherwise write flash every few milliseconds. The two CAPTURE buttons save on their own; the offsets do not.

8.9 COLORS tab

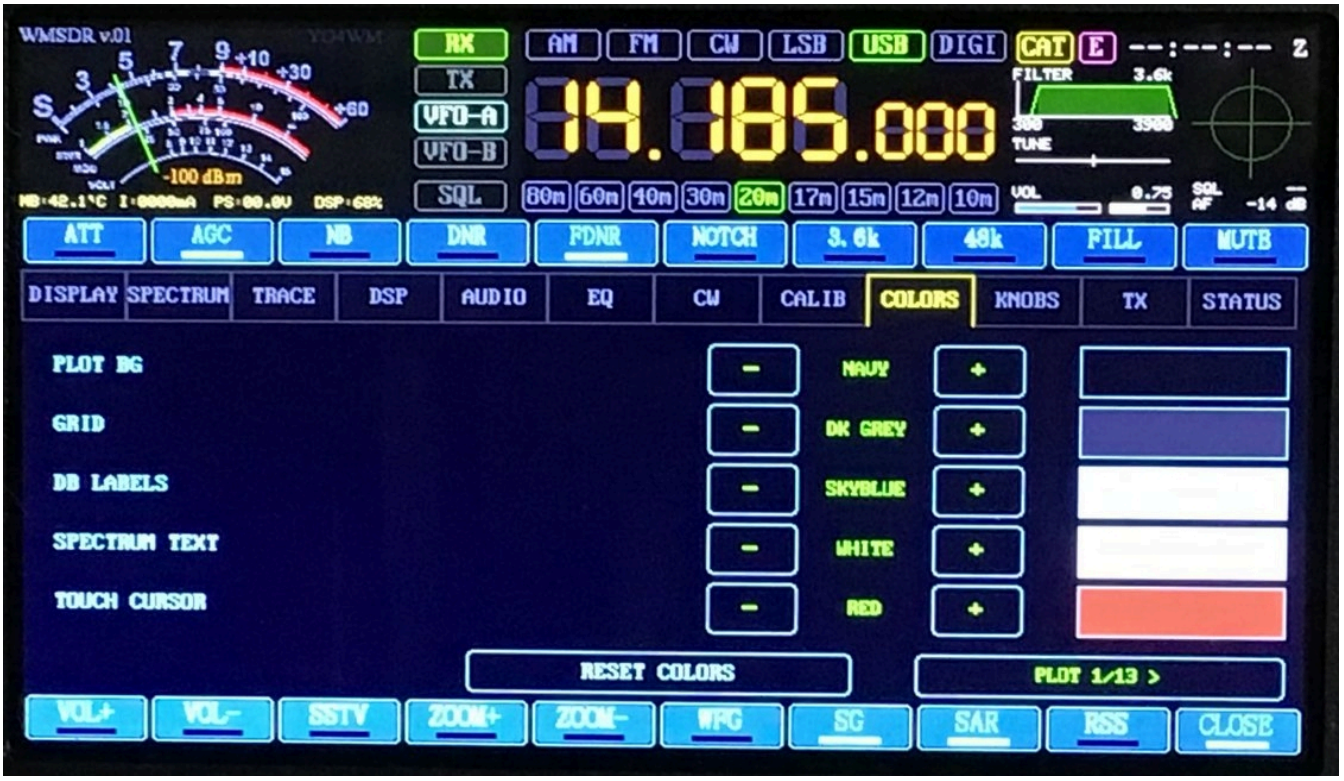


Figure 8.9 — COLORS tab.

Every colour in the UI is a themeable slot. There are more slots than fit one page, so the tab shows one **group** at a time and the selector on the action line steps through them.

Each row shows the slot name, `-` / `+` to walk the palette, and a **swatch** — several palette entries are hard to tell apart by name on a dim panel, and the swatch is the only honest preview for a slot whose value is off-palette (shown as hex). The palette wraps, so there is no end of travel.

Where you left the group selector is not persisted; the colours themselves are.

Action: RESET COLORS. Also load-bearing: the palette contains BLACK, so it is entirely possible to set a text slot to its own background and lose the ability to read the tab you would need to undo it. This resets **every** slot, including those on other groups.

8.10 KNOBS tab



Figure 8.10 — KNOBS tab.

Assigns the optional **8Encoder** knob unit (§2). Without the unit fitted the tab still works and the settings are kept, ready for when it is plugged in.

How the knobs behave

- **Turning** a knob moves one setting by **one menu step per click** — exactly as the `-` / `+` buttons of that setting's own tab would, with the same limits. With that tab open, the value on screen follows the knob.
- Each knob has up to **four pages**. A page is one setting plus an **LED colour**; the knob always drives its current page, and its LED shows which page that is. This is how one knob can cover a whole group — the default AGC knob is `AGC TARGET` (red), `AGC HANG` (green) and `AGC RELEASE` (blue).
- Each knob's push button knows two presses: **short** (released before 0.6 s) and **long** (fires at 0.6 s, while still held — you do not need to let go). The LED flashes **white** on a short press and **orange** on a long one, so you can see a press was taken.
- The **toggle switch** on the unit arms or inhibits all eight knobs. Inhibited, the knob LEDs go dark and turns and presses are ignored — use it so a stray hand cannot change a setting mid-QSO.
- Pages always start on **page 1** after power-on.

The tab, top to bottom

Area	What it does
1 … 8	Selects the knob to edit. The bar under each number is that knob's live LED colour. **Turning or pressing a physical knob while this tab is open selects it here** — and does nothing else, so setting up cannot change a value by accident.
LED − / +	Master brightness of all the unit's LEDs, **5 – 100 %**, step 5, default **30 %**. The steps are perceptual, not linear, so the low end is usable at night. The minimum is not 0: the switch is how the LEDs go dark.
PAGE 1 … 4	`<` / `>` on the left walk the list of settings: `- none -`, then every numeric setting on the SPECTRUM, TRACE, DSP, AUDIO, EQ and CW tabs, shown as `TAB: SETTING`. Press and hold to scroll. `<` / `>` on the right pick the LED colour: RED, ORANGE, YELLOW, GREEN, CYAN, BLUE, PURPLE, MAGENTA, WHITE. The live page is marked `*`.
SHORT < >	What a short press does: **NONE**, **NEXT PAGE** or **PREV PAGE**.
LONG < >	The same choice for a long press.

Editing a page makes it the knob's current page, so the knob's LED on the unit previews the colour as you step through it.

A page showing `? MISSING` points at a setting a firmware update renamed or removed. It does nothing; pick another setting for it.

CALIB settings cannot be put on a knob: they are only kept through **SAVE CALIBRATION**, so a knob there would move a value that silently reverts at the next power cycle.

Defaults

Knob	Pages	Short press
1	VOLUME (green)	—
2	AGC TARGET (red) · AGC HANG (green) · AGC RELEASE (blue)	NEXT PAGE
3	NB THRESHOLD (orange)	—
4	DNR LEVEL (magenta)	—
5	FDNR FLOOR (purple)	—
6	FM SQUELCH (yellow)	—
7	CW TONE (red)	—
8	WF CONTRAST (cyan)	—

Saving. Knob assignments and LED brightness are saved like every other setting — on menu close or by the 3-second autosave — but as their own records, so a firmware update that changes the main settings record does not lose your knob setup (and the reverse).

Action: RESET KNOBS. Restores the default assignments above. It touches nothing else — not the values the knobs control, not the LED brightness.

8.11 TX tab



Figure 8.11 — TX tab.

Transmit settings. **Nothing transmits while TX ARM is OFF**, whatever asks for it.

Row	Range	What it does
TX ARM	OFF / ARMED	Allows transmitting. **Never saved — every boot starts OFF.** While armed the TX lamp has an orange outline (§6.2). Switching OFF drops PTT at once and stops a CW over.
CW WPM	5–60	Keyer speed (PARIS). Also set from the browser Transmit page.
PTT TIMEOUT	5–240 s	The longest a single key-down may last; PTT is then forced off (logged as `PTT TIMEOUT`). In CW every dit and dah is its own key-down. Voice overs use VOICE TIMEOUT instead.
VOICE TIMEOUT	30–300 s, step 10	The longest voice over. When it is reached the over ends completely — PTT off, mic off, receive audio back (logged as `VOICE TIMEOUT`). Holding the mic PTT does not re-key: release it and press again.
CW WEIGHT	25–75 %	Mark length against the gap after it. 50 % is standard; heavier sounds fuller and carries better on a weak or fading path, lighter sounds crisper. Speed is unchanged.
DAH RATIO	2.5–4.5	Dah length in dits (standard 3.0).
LETTER SPACE	3–6 units	Gap between letters (standard 3).
WORD SPACE	5–14 units	Gap between words (standard 7).
SIDETONE	0–10	CW monitor level in the BEESDR audio; 0 = off.
SIDETONE PITCH	300–1200 Hz	Sidetone pitch — its own setting, independent of TONE on the CW tab.
MIC BOOST	OFF / +20 dB	The codec's microphone preamp boost. Takes effect at the next mic PTT press.
MIC GAIN	0–40 dB	Digital gain after the converter: the level going into the speech processor. Watch the **MIC** and **COMP** bars (§6.9).

Row	Range	What it does
TX LEVEL	−70–0 dB	The level sent to the uSDX mic input. Set it with the 1 kHz tone (§9, *Sending voice*).
TX HICUT	2400–3000 Hz, step 100	Top of the transmitted audio band — match the uSDX SSB filter. Shown as the right edge of the TX AUDIO mask.
TX TONE	MIC / 1 kHz / 2-TONE	Test tones in place of the mic, for setting levels. 2-TONE is 700 + 1900 Hz at the same peak level as the single tone. Tones bypass the MIC tab processing. **Never saved — every boot transmits the mic.**
TX OUTPUT	L+R / L ONLY / R ONLY	Which output channels carry the TX audio. L ONLY keeps the right channel silent while transmitting, R ONLY the left; the uSDX branch must then be on the channel left live (L = the tip, R = the ring).

Everything except TX ARM and TX TONE is saved, in its own record (a menu RESET DEFAULTS does not touch it).

PTT. The line is the AW9523 expander pin P0_0 on the codec board, wired to the uSDX PTT: driven **HIGH at standby, LOW while transmitting**. Connect nothing else to that line — the pin actively drives it. The **mic PTT button** is a separate input, pin P0_1 (switch to ground, with a pull-up on the board).

Voice transmit. Pressing the mic PTT mutes receive audio, switches the codec input to the mic and keys P0_0 as soon as the output is silent at the jack (a few tens of milliseconds), so no receive audio reaches the uSDX. Releasing fades the audio to silence first, then unkeys. The press is refused while TX ARM is OFF and ignored while the CW keyer is sending. The audio passes through MIC GAIN, a 250 Hz high-pass, the speech processor on the MIC tab (§8.12), a final band-pass and an output limiter, then TX LEVEL. The band-pass limits the transmitted audio to **300 Hz – TX HICUT** with steep edges (−6 dB on the two mask lines, about −74 dB within 150 Hz outside them), so nothing the processor generates leaves the band. While transmitting, the spectrum strip shows the TX monitor (§6.9).

CW transmit. Text comes from the browser Transmit page (§7.4). The keyer times each element on a microsecond timer and keys PTT directly, so element lengths stay exact to at least 45 WPM. The uSDX, in CW mode, makes the carrier; no audio is involved.

CW paddle. A paddle can be wired to the codec board: **DIT on P0_6, DAH on P0_7**, contacts to ground. Each line needs a **pull-up resistor to 3.3 V** on the board. Without them the inputs float, and after a few seconds the keyer starts sending on its own. The keyer is **iambic A** at CW WPM, with the same CW WEIGHT, DAH RATIO and spacing as typed text, and the same sidetone. It needs **TX ARM** and the **rig in CW** (the mode read over CAT); otherwise the paddle is ignored. Receive audio is muted from the first element until one word space (7 dits) after the last, then it comes back. It works in the normal radio and in Zeus mode (§7.5). If DIT and DAH come out swapped, swap the two wires.

Sidetone. Heard only in **CW mode while the keyer sends**. It replays the real PTT edges, sample-accurate, with a small fixed delay, and **replaces receive audio** for the over; receive audio comes back when the over ends. The BEESDR audio output also feeds the uSDX mic input, but in CW mode the uSDX ignores the mic.

> **uSDX VOX must be OFF** once BEESDR's audio output is patched to the uSDX mic: that output carries
> receive audio between overs, and VOX would key the rig on it.

8.12 MIC tab

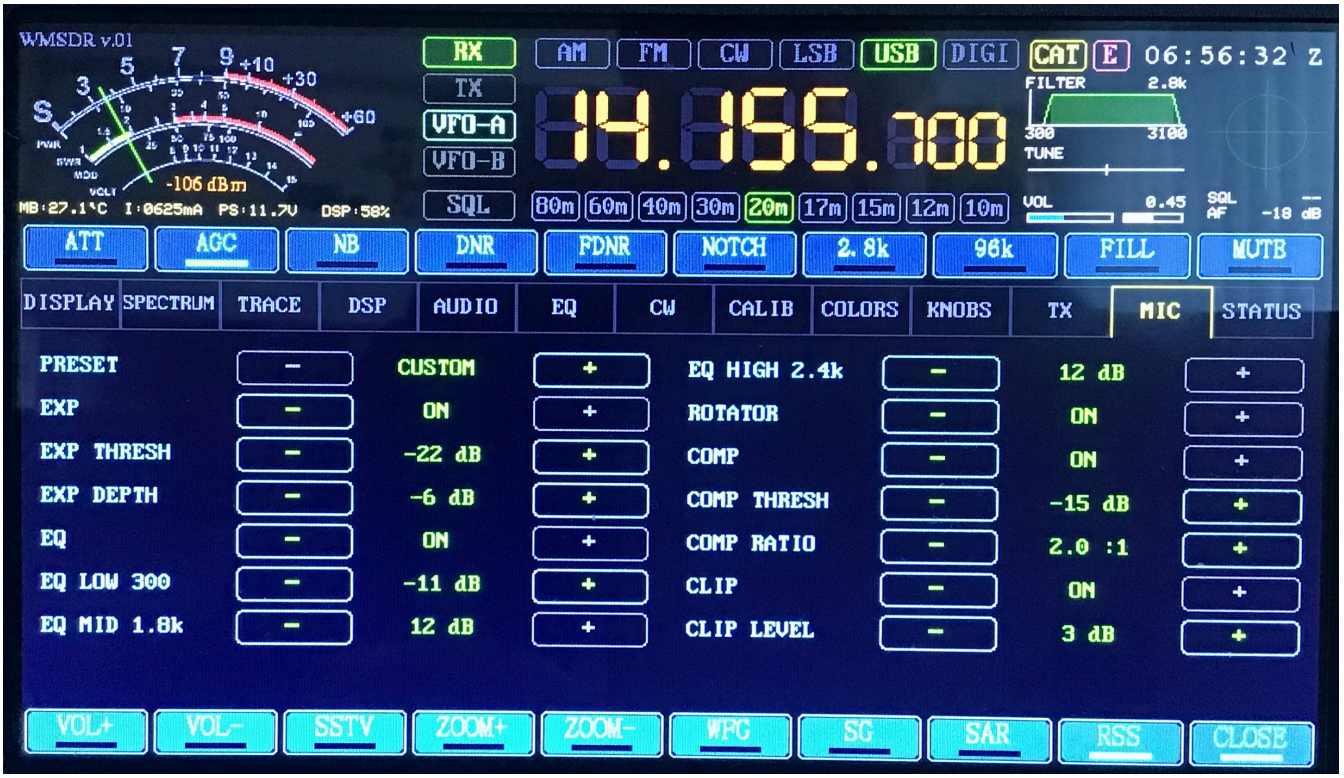


Figure 8.12 — MIC tab.

The speech processor for voice transmit. The rows are in chain order — EXP, EQ, ROTATOR, COMP, CLIP — each stage with its own ON/OFF so it can be compared on the air. Every change applies mid-over. The TX monitor bars (§6.9) show what each stage is doing.

> **Top-tier processing on a microcontroller.** This is the same family of building blocks found in the transmit chains of high-end SDR transceivers and their PC software: downward expander, multi-band EQ, phase rotator, compressor, and an envelope clipper that filters while the signal is still complex so the clipping products stay out of the passband — followed by a look-ahead limiter and a live spectrum and level display of what goes on the air. Here it all runs in real time on the same ESP32-S3 that runs the receiver, and turns a plain uSDX into a rig with a modern speech processor in front of it.

Row	Range	What it does
PRESET	CUSTOM / RAGCHEW / DX	Sets every row below in one step (MIC GAIN, MIC BOOST, TX LEVEL and TX HICUT on the TX tab are left alone). Changing any row afterwards turns it back to CUSTOM.
EXP	OFF / ON	Expander: a soft noise gate that lowers the room noise between words.
EXP THRESH	−60 … −10 dB	Below this level the expander starts to attenuate. **Keep it below COMP THRESH**, or normal speech gets cut between syllables.
EXP DEPTH	−20 … 0 dB	The most it attenuates. Too deep makes the background switch on and off with every pause, which sounds broken up; −6 dB sounded right on the air.
EQ	OFF / ON	Transmit equaliser, three bands.
EQ LOW 300	−12 … +12 dB	Low shelf at 300 Hz. Cutting it removes boom and saves power for the frequencies that carry speech.
EQ MID 1.8k	−12 … +12 dB	Presence peak at 1.8 kHz.
EQ HIGH 2.4k	−12 … +12 dB	High shelf at 2.4 kHz.

Row	Range	What it does
ROTATOR	OFF / ON	Phase rotator: makes the voice waveform more symmetric so its peaks are lower. No audible change on its own; it lets the clipper work harder before it sounds harsh.
COMP	OFF / ON	Compressor: brings quiet speech up towards the loud parts.
COMP THRESH	−40 … 0 dB	Level above which it compresses.
COMP RATIO	1.5 … 6.0 : 1, step 0.5	How hard it compresses above the threshold.
CLIP	OFF / ON	Clipper: cuts the envelope peaks, then filters the result while the signal is still complex, so the clipping products stay out of the passband. The main source of talk power.
CLIP LEVEL	0 … 12 dB	How hard the signal is driven into the clipper. The output level stays the same; more drive means more clipping.

The presets:

	EXP	EQ low / mid / high	ROTATOR	COMP	CLIP
RAGCHEW	ON, −22 / −6 dB	ON, −3 / +2 / 0 dB	ON	ON, −15 dB, 2 : 1	OFF
DX	ON, −22 / −6 dB	ON, −6 / +4 / +2 dB	ON	ON, −20 dB, 3 : 1	ON, 6 dB

RAGCHEW keeps the voice natural; DX is punchier, with less bass and more clipping. On the air with an electret mic, **MIC BOOST +20 dB and MIC GAIN 20 dB** put the compressor in its working range (COMP bar about 6–12 dB on normal speech). Without the rotator, clipping started to sound harsh at CLIP LEVEL 6; with it, 6 was clean.

Saved with the TX tab's record. This tab has no action button.

8.13 STATUS tab

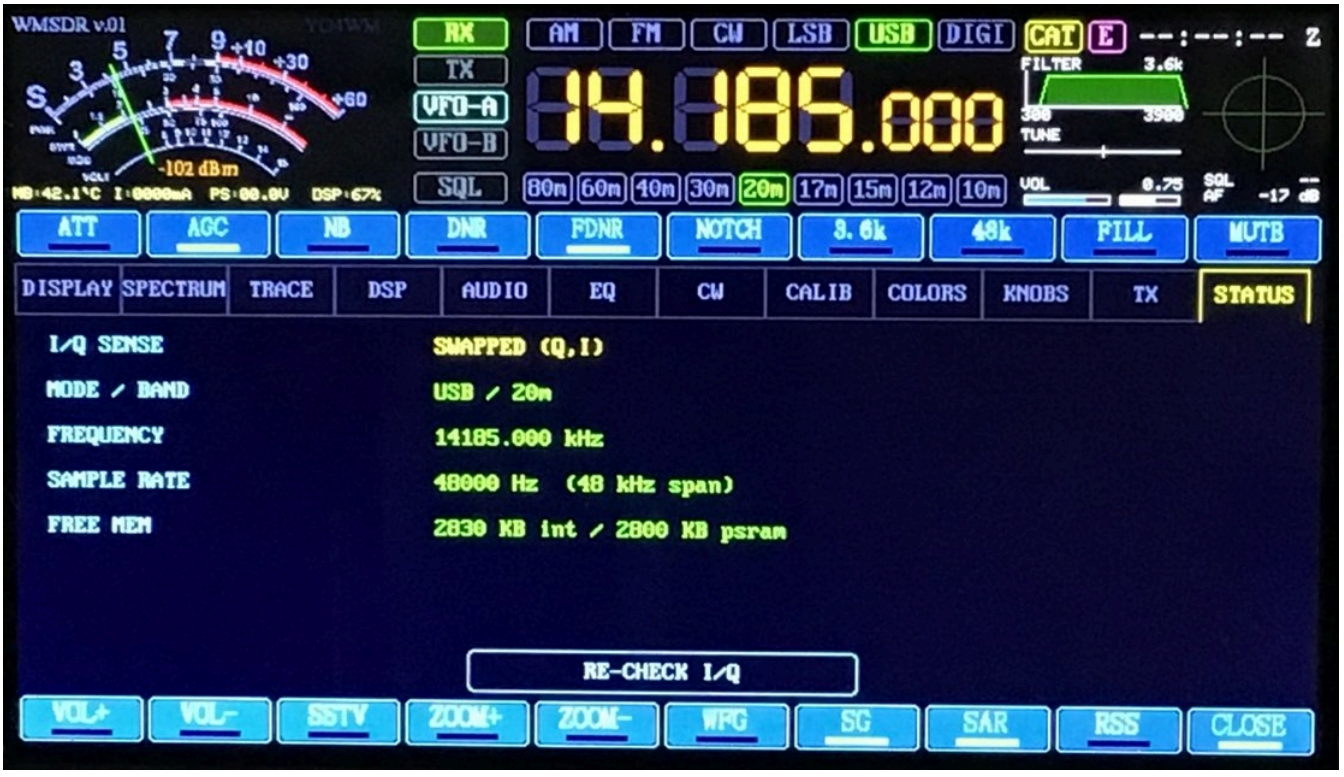


Figure 8.13 — STATUS tab.

Read-only diagnostics, refreshed twice a second.

Row	What it tells you
I/Q SENSE	`STRAIGHT (I,Q)` (green) or `SWAPPED (Q,I)` (yellow), or `checking...` while detection runs. Either result is fine — BEESDR compensates. This is the fastest confirmation that the I/Q pair is actually arriving.
MODE / BAND	What the rig says, via CAT
FREQUENCY	The current LO, in kHz
SAMPLE RATE	The current rate in Hz, and the resulting span
FREE MEM	Free internal DRAM and free PSRAM, in KB

Action: RE-CHECK I/Q. Re-arms the automatic sense detection. It runs from inside the DSP frame, where the buffers it inspects are actually valid, so the result appears a moment after the press. Use it after a mode change, a band change, or any time the sideband looks reversed.

Nothing on this tab is persisted — a known state at boot is more useful here than a remembered one.

---

9. Practical setup sequences

Getting on the air the first time

1. `CAT BAUD` (DSP tab) to match the rig · check STATUS shows the frequency moving.
2. **RE-CHECK I/Q** on STATUS. Confirm on the air: tune the rig **down**, the carrier should move **right**.
3. `I/Q` button → pick a span. 48k is the general-purpose choice.
4. CALIB tab → CAPTURE NOISE at S1, CAPTURE SIGNAL at S9, **SAVE CALIBRATION**.
5. `FILTER` → 2.4k for SSB.
6. `AGC` on, `VOLUME` to taste — keep the `AF` readout around −6 dB on voice.

The band is noisy

1. `NB` on. Watch the `NB <n>` count in the DSP log while setting `NB THRESHOLD` — near zero on a clean band is what you want.
2. Try `DNR` and `FDNR` **one at a time** on the same signal. They are different tools.
3. If FDNR burbles — a faint "radio station" in the gaps between words — check `FDNR ALGO` is on **HYBRID** first; MIN-STAT does this by design. If it burbles on HYBRID, raise `FDNR FLOOR` towards −3.
4. A steady het: `NOTCH`.

The waterfall looks wrong

1. Set `WF FLOOR BLEND` to **0**. That collapses the per-bin scheme to a single global floor — roughly the classic look. If it looks right now, the per-bin tracking was the cause and `WF BLACK LEVEL` / `WF CONTRAST` are where to work.
2. Remember the menu suppresses the waterfall repaint — close it and wait for the old history to scroll off before judging anything.

The display has gone unreadable

Open the menu, find the tab that owns the setting, press **RESET DEFAULTS** (or **RESET COLORS** on the COLORS tab). A power cycle will *not* rescue you — these values are persisted.

Setting up a knob

1. Check the unit's switch is on — the knob LEDs are lit.
2. Menu → **KNOBS**. Turn the physical knob you want; its number is selected on screen.
3. **PAGE 1** → `<` / `>` to the setting, then pick a colour.
4. More than one setting on this knob? Fill **PAGE 2** onwards, and set **SHORT** (or **LONG**) to **NEXT PAGE**.
5. Close the menu and try it. Too bright or too dim? **LED – / +** on the same tab.

Receiving an SSTV picture

1. Tune as for voice on an SSTV frequency (§7.3) — USB on 20 m, LSB on 40 / 80 m.
2. **SSTV** button → MODE **AUTO**, HOLD off.
3. **TUNE**: wait for the needle to rest on 1900 — a header is coming. Press TUNE again to watch the picture.
4. Nothing after the 1900? Press **START** as the picture begins.
5. **HOLD** to keep a finished picture.

Receiving a weather chart

1. Tune DWD in USB, 1.9 kHz below the carrier (7878.1 kHz for 7880).
2. **SSTV** button → MODE **WEFAX**.
3. Wait for `start signal detected` and the phasing — or, mid-chart, press **START** and tap the chart's left margin.
4. Lines leaning? **SLANT±** first, then tap the margin again if needed.

Receiving RTTY

1. Tune USB so the tones sit on the **M** / **S** markers of TUNE (DWD: dial 10099.1 kHz for 10100.8, or 11037.3 kHz).
2. **SSTV** button → MODE **RTTY**; tap the status line for the preset (DWD **50/450**, amateur **45.45/170**).
3. Garbage on a clean signal? **REV**.
4. Let the AFC settle; **CLEAR** empties the console, **HOLD** freezes it.

Receiving FT8

1. Check the **E** icon is lit — FT8 needs the companion's UTC.
2. Tune **14.074 MHz USB** (or another FT8 frequency, §7.3).
3. **SSTV** button → MODE **FT8**. Decodes appear after the first full 15 s slot.
4. To read at leisure: **HOLD** on the radio, or open `beesdr-dx.local/ft8.html` and use **Pause**, the filters and **Save log** (§7.4).

Sending CW

1. uSDX on a dummy load or the antenna, **CW mode**, **VOX off**.
2. MENU → **TX** → **TX ARM** → **ARMED**. Set CW WPM and SIDETONE. Close the menu — the TX lamp is outlined orange.
3. Open `beesdr-dx.local` → ☰ → **Transmit**. The badge shows **ARMED**.
4. Type — each character goes out as you type it. Macros for CQ / RST / 73. **STOP** or **Esc** stops at once.

5. Done: **TX ARM → OFF**.

Sending voice (SSB)

Once, for the cable. The BEESDR audio output goes to the uSDX mic input (L, the tip, if you want TX OUTPUT = L ONLY; on R, the ring, for R ONLY). The uSDX puts about **5 V of electret bias** on its mic tip: fit a **DC-blocking capacitor** in the uSDX branch (4.7 µF, **+ towards the uSDX**). A resistive attenuator in the same branch is recommended (10 kΩ in series, 100 Ω to ground at the uSDX end, about −40 dB) — the BEESDR output is much hotter than a mic, and without it TX LEVEL has to sit near the bottom of its range.

1. uSDX in **USB or LSB**, **VOX off**, into a dummy load; a second receiver or WebSDR listening.
2. MENU → **TX** → **TX ARM → ARMED**, **TX TONE → 1 kHz**, TX LEVEL low.
3. Press the mic PTT and raise **TX LEVEL** until the uSDX output power stops rising, then back off about 3 dB. Check **2-TONE**: two clean lines on the second receiver, nothing either side.
4. **TX TONE → MIC.** MIC BOOST +20 dB, then MIC GAIN so that normal speech gives about 6–12 dB on the **COMP** bar (§6.9). If COMP stays near 0, raise MIC GAIN; if it sits above 15, lower it.
5. MENU → **MIC** → **PRESET → RAGCHEW** or **DX**. Speak, and watch the TX monitor: speech inside the shaded band, **CLIP** mostly under 20 %, **OUT** touching red only on the loudest syllables.
6. Done: **TX ARM → OFF**.

Using a PC program (Zeus / Thetis / piHPSDR)

Once. Voice levels set as in *Sending voice* above, in the normal radio; Zeus mode uses the same settings. A firmware built with your Wi-Fi network (§7.5).

1. MENU → **DSP** → **ZEUS → ON**. The BEESDR reboots into the Zeus screen and shows its IP address, then `waiting for client`.
2. On the PC, start the program and choose the radio it finds. Set **48 kHz**. In Zeus, also **Settings → Radio → Audio output = "Radio Speaker/headphone"**.
3. Start the program. The screen shows `STREAMING`, EP6 and EP2 both near 381/s; the rig moves to the program's frequency. Set the rig's **mode by hand** to match.
4. Voice: the program's **MOX**, or the hand-mic PTT. CW: rig in **CW**, then the **paddle**.
5. Done: hold **EXIT** for 1 second. The BEESDR reboots into the normal radio.

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10. Annexes

Annex A — Interface board schematic

To be inserted.

Covers: I/Q input conditioning and level setting into the WM8731 line inputs, CAT level translation between the rig and the ESP32-S3 UART, the audio output stage, and the supply arrangement.

Annex B — Proposed enclosure



Figure B.1 — Enclosure front panel: touch panel, eight-encoder unit with LED indicators, tuning knob, ATU display (separate unit), round multi-pin connector bottom left. See also Figures 1 and 2.

Dimensioned drawings: *to be inserted.*

Annex C — Reference tables

To be inserted.

Suggested: pinout summary, default values for every menu row, and a one-page quick reference card for the two button rows.

Annex D — Signal processing block diagrams

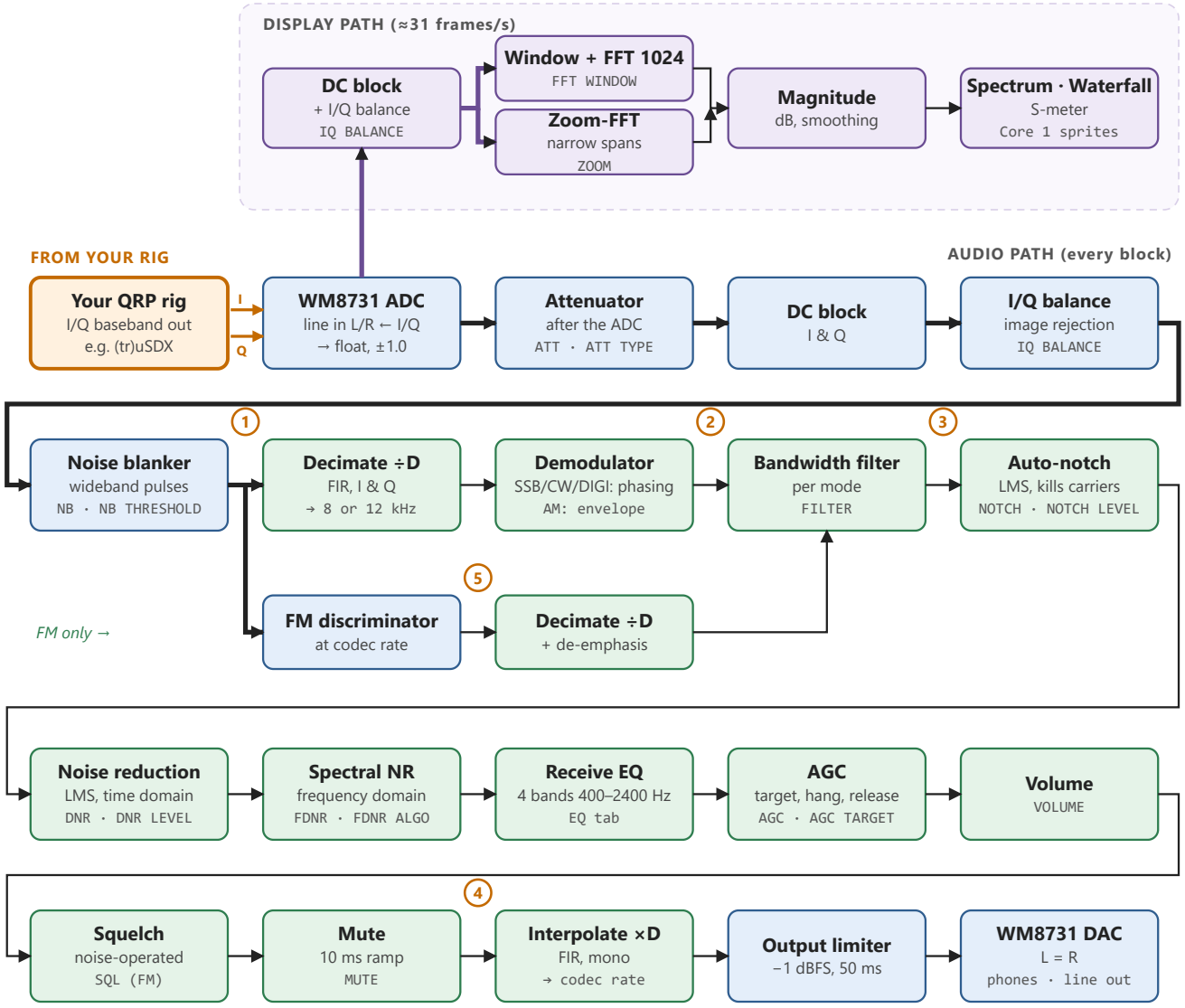
BEESDR never touches RF. Your rig does the RF work and brings the band down to an I/Q baseband. BEESDR digitises that I/Q in the WM8731 codec, and everything after that is software running on core 0 of the ESP32-S3, one block of 1024 samples at a time (21.3 ms at 48 kHz). The two figures below show that software stage by stage, in the order the firmware runs it.

How to read them:

- **Colour = sample rate.** Orange boxes are external equipment, blue boxes run at the codec rate (8, 32, 48 or 96 kHz, set by the I/Q span), and green boxes run at the audio rate (8 or 12 kHz). Purple boxes are the display. Decimation (÷D) and interpolation (×D) are where the rate changes.
- **Arrows.** A thick arrow carries the I/Q pair, a thin arrow carries mono audio, and a dashed arrow is PTT or control, not audio.
- **Grey monospace text** inside a box is the menu row or button that controls that stage (see §7 and §8).
- **Numbered circles are taps.** A tap is a read-only copy of the signal at that point, used by a decoder or a meter. The chain itself is not changed by it.

BEESDR — receive signal chain

Core 0, processing_task: every 1024-sample block (21.3 ms at 48 kHz) goes through both paths



Legend

- external rig — analogue I and Q
- codec rate (8 · 32 · 48 · 96 kHz)
- audio rate (8 or 12 kHz)
- display
- I/Q pair (complex)
- audio (mono)
- ABC MENU / BUTTON name

Taps (read-only copies, the chain is not changed)

1. CW decoder — I channel at codec rate, ahead of all filtering (CW / CWR)
2. CW tuning aid — before the bandwidth filter (CW / CWR)
3. Decoder window — SSTV · WeFax · RTTY · PSK31 · FT8
4. AF level meter — exactly what reaches the DAC
5. FM noise metric — drives the squelch (SQL)

BEESDR never touches RF: it processes the I/Q baseband your rig already produces.

D = 1 · 4 · 4 · 8 for the 8 · 32 · 48 · 96 kHz spans. During a voice over both paths are skipped and the TX chain runs instead.

Figure D.1 — Receive chain. The same block from the codec feeds two paths: the display path (top, about 31 frames per second) and the audio path (every block). FM leaves the main path after the noise blanker and joins it again at the bandwidth filter. Squelch and MUTE sit after the AGC on purpose, so the AGC never winds up while the audio is gated.

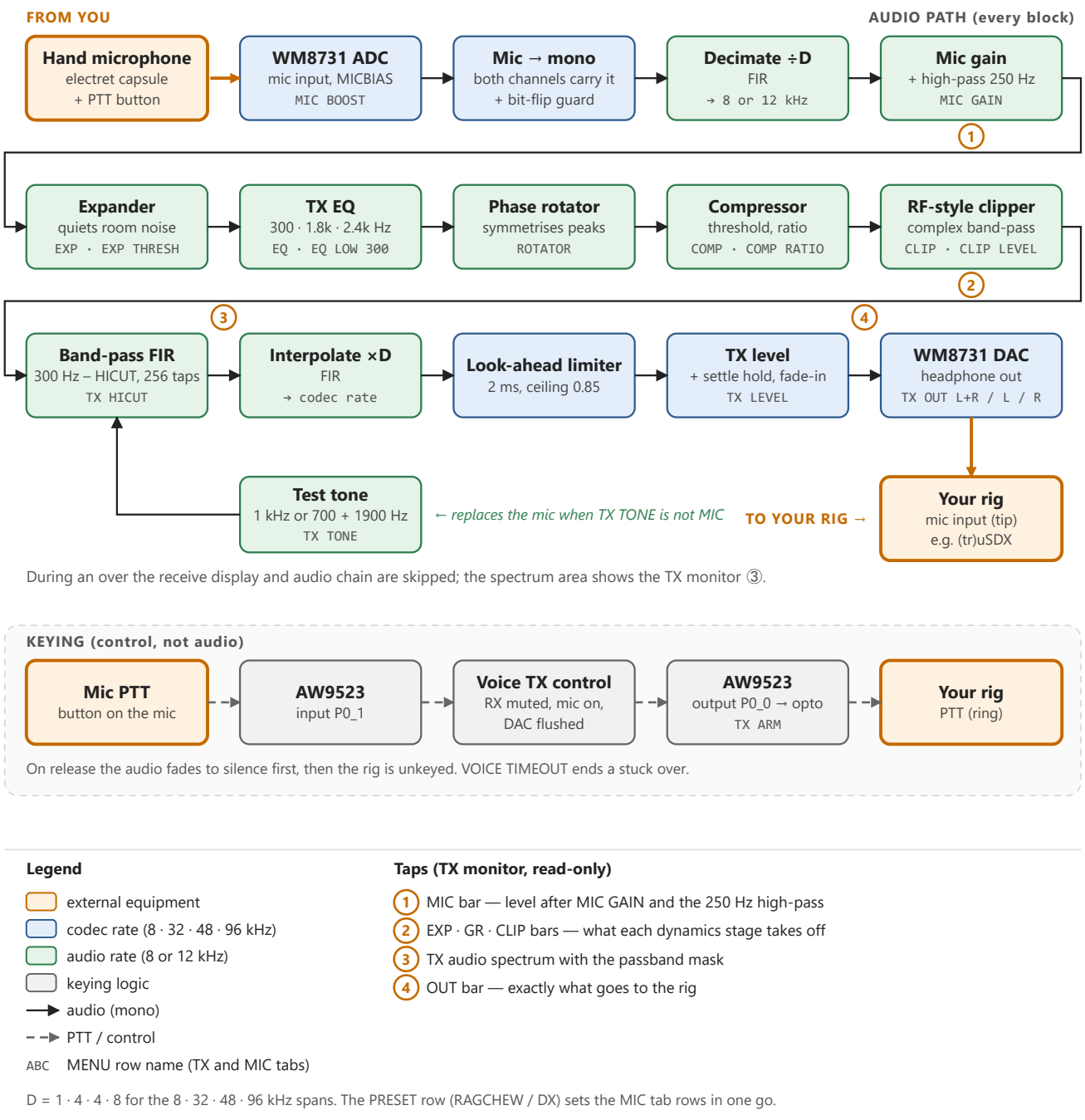


Figure D.2 — Voice transmit chain. While the mic PTT is held, this chain replaces the receive chain on the same core, and the spectrum area shows the TX monitor (§6.9). The test tone (TX TONE) replaces the microphone and skips the dynamics stages. The keying band at the bottom shows how the mic's PTT button keys the rig; on release the audio fades to silence before the rig is unkeyed.
